# Where we are...

Assignment 1 is due this week; you can use AI in whatever way you want

Please finish your Labs 4 & 5 this week

Lab 6 (next Wed) is an in-class activity; after that 🌸

An extra credit assignment will be released this week; due May 6 2026

# Animation

## CMSC471: Introduction to Information Visualization

Fumeng Yang, Assistant Professor

March 2, 2026

University of Maryland, College Park

https://fumeng-yang.github.io/CMSC471

# Animation

Interaction

## Likes

Understand a state transition

Direct attention

Visual variable to encode data

Explain a process

Increase engagement/entertainment

Storytelling

## Struggles

Cognitive overload

Distracting/confusing

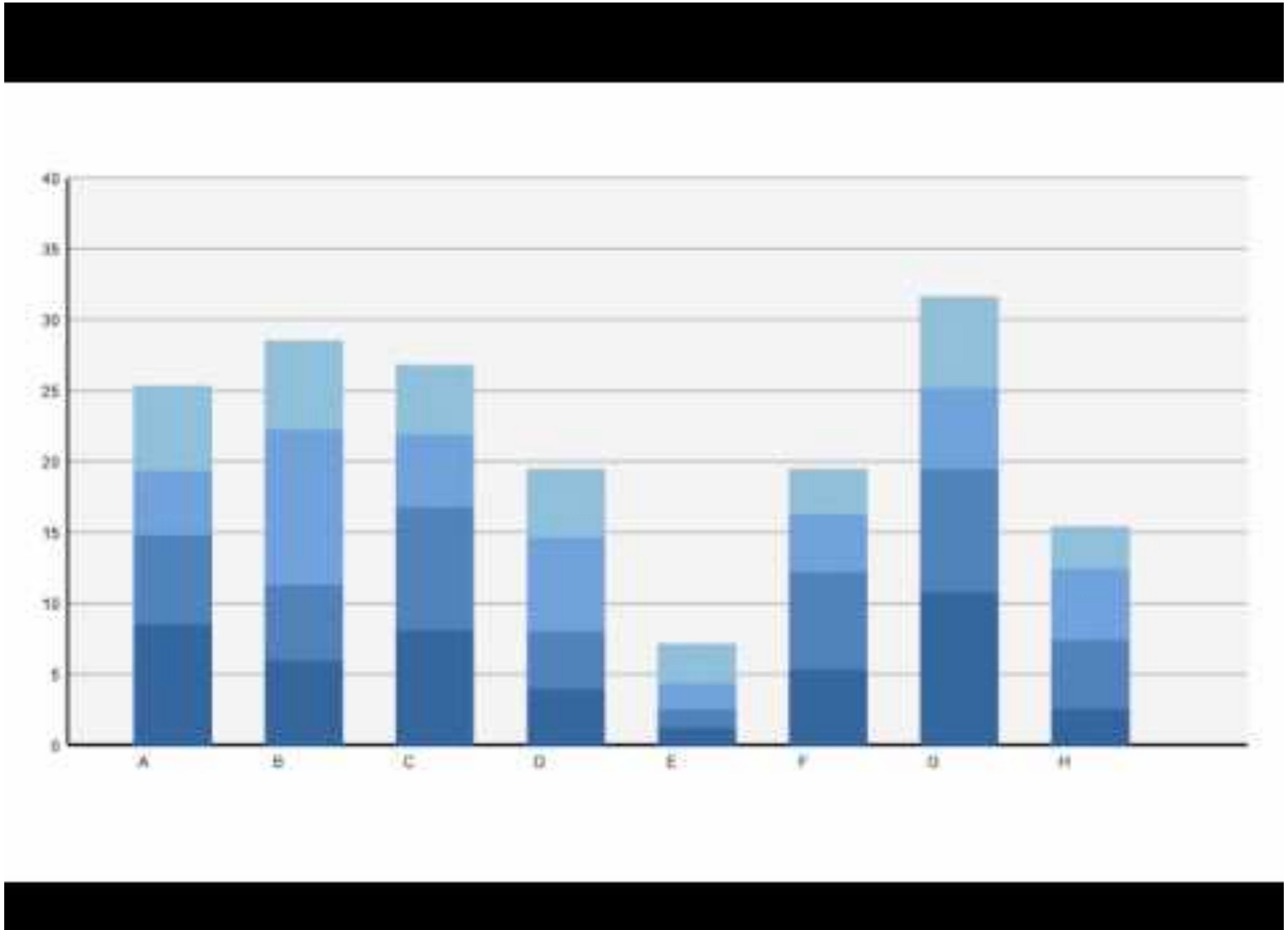
Perceived as unscientific

Tedious

Implementation/performance

# Animations we like

# Understand a state transition




https://www.youtube.com/watch?v=vLk7mlAtEXI

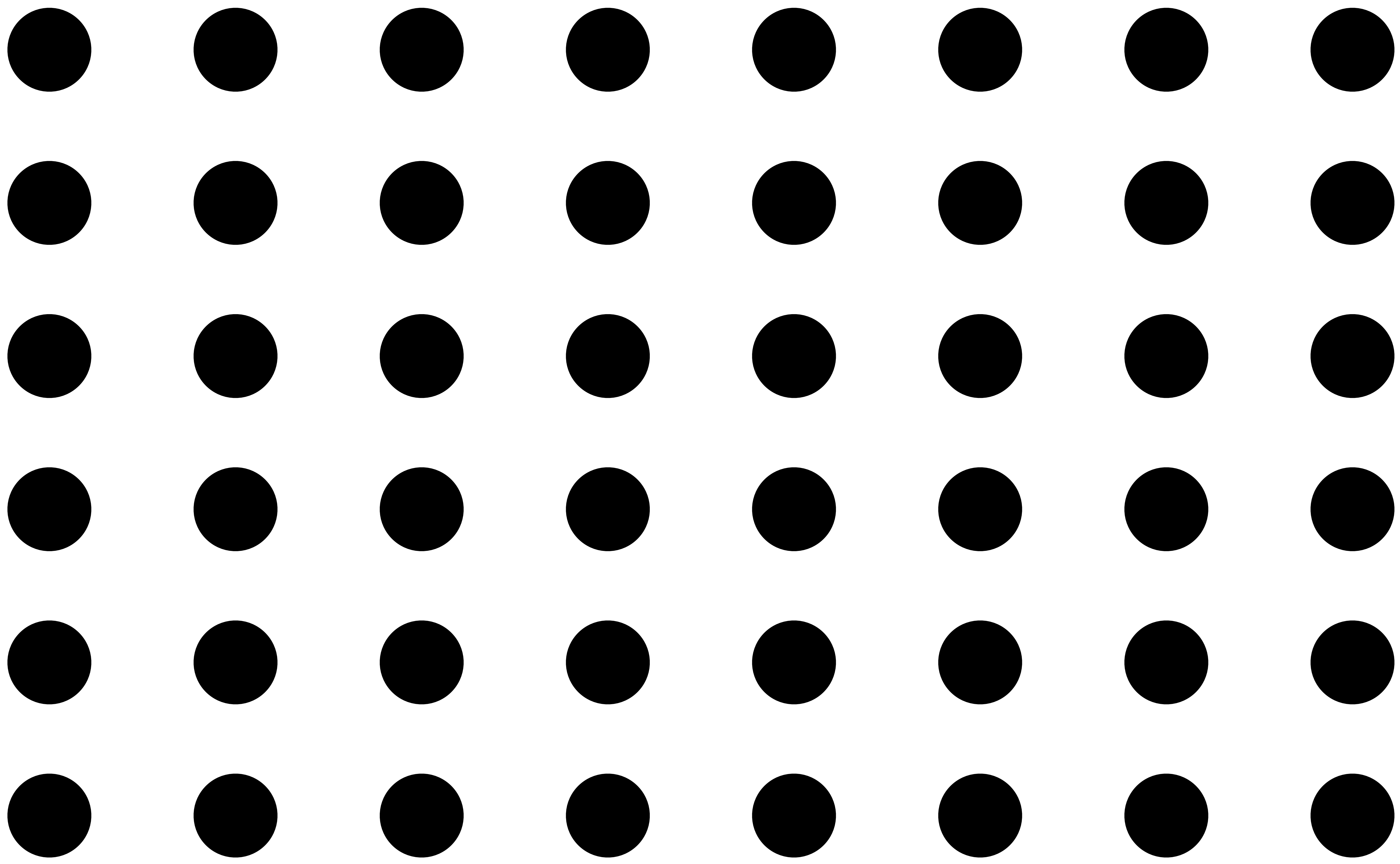
All can be implemented in D3.js

# Direct attention

**Motion as a visual cue**

Pre-attentive, stronger than color, shape, etc.

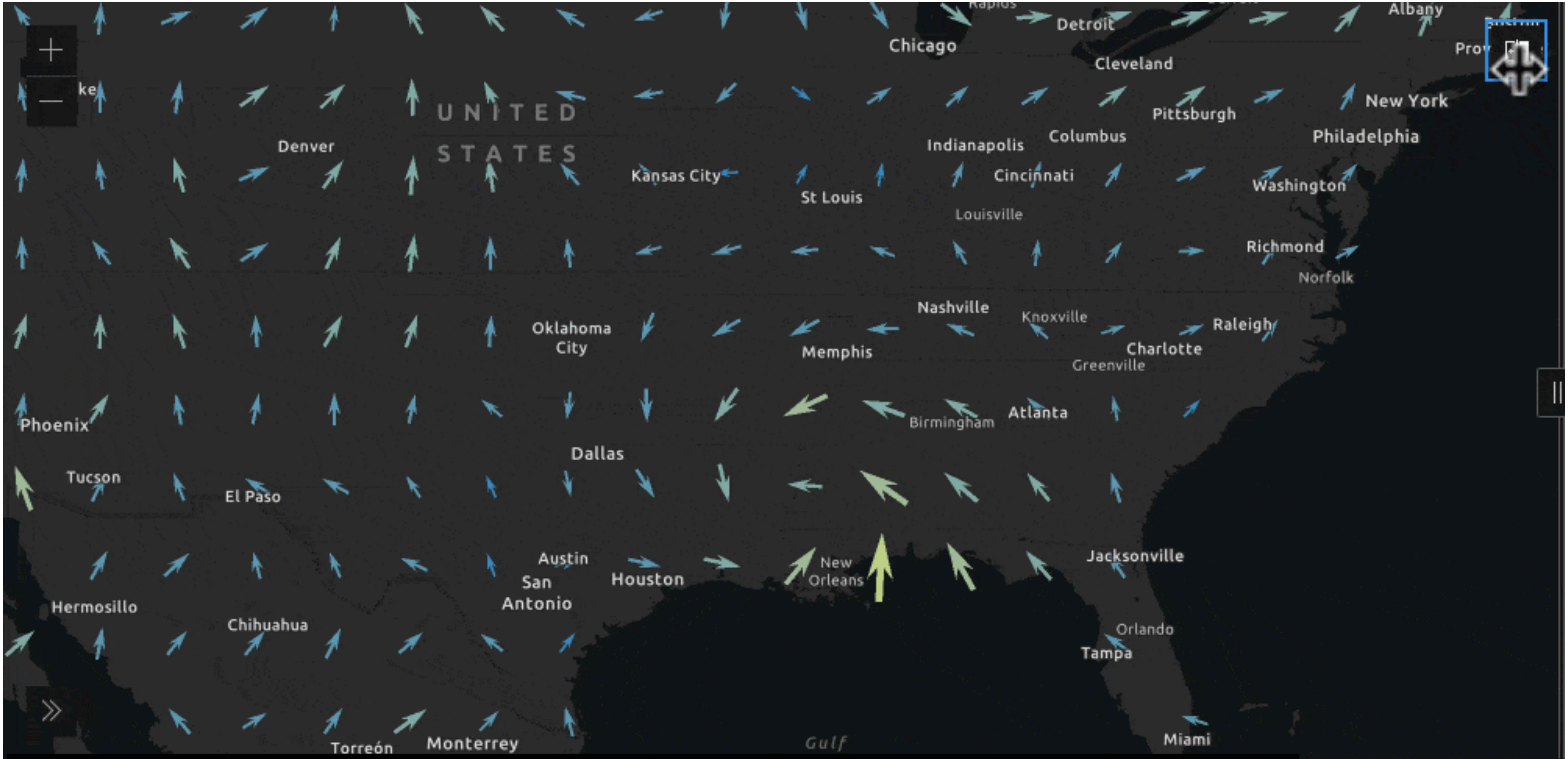
I use it a lot in my slides to guide your attention



# Variable


http://hint.fm/wind/

# … probably better than this

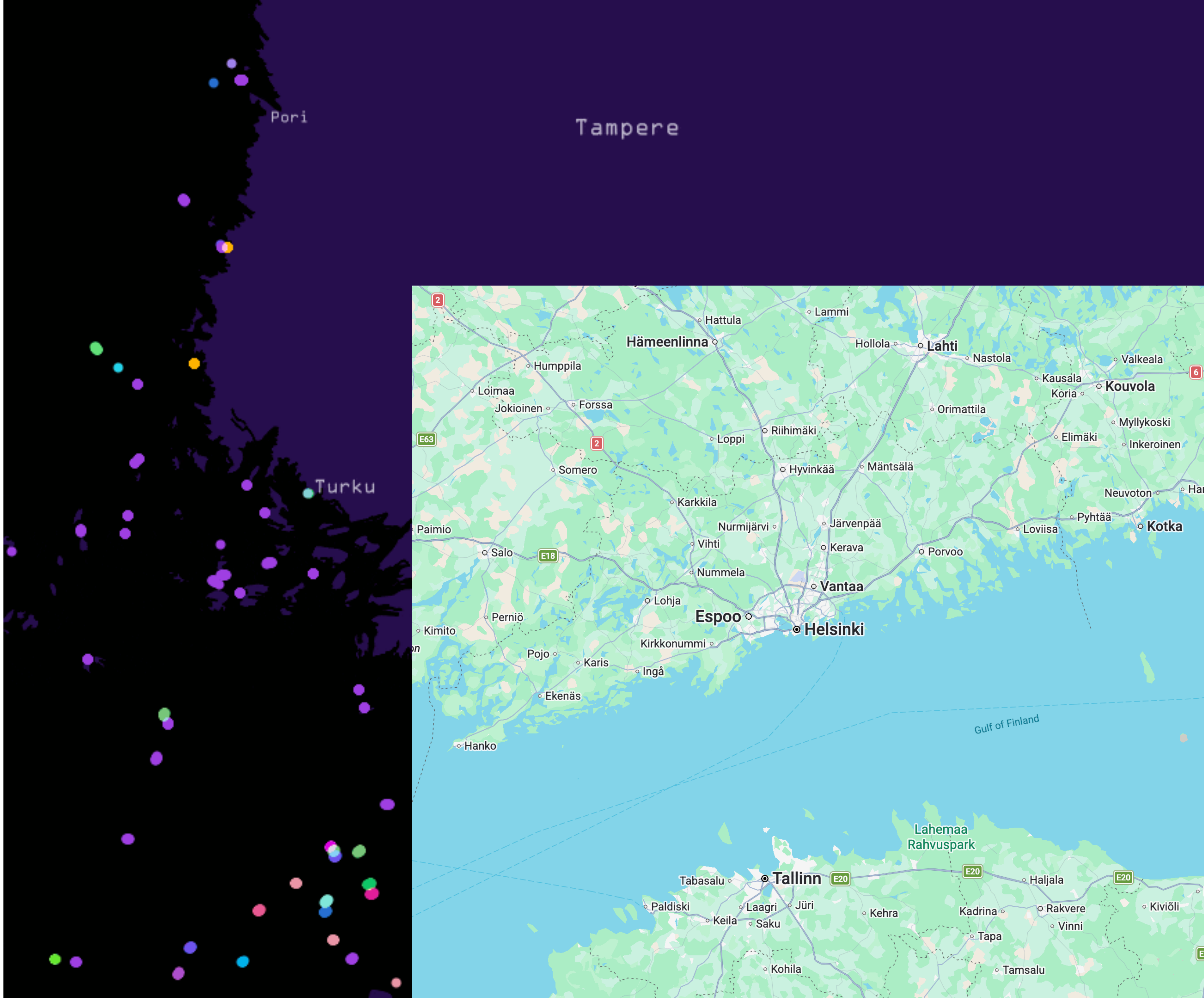



https://www.esri.com/arcgis-blog/products/js-api-arcgis/mapping/create-an-animated-flow-visualization-with-the-arcgis-api-for-javascript/

# Variable

24 hours of marine traffic on the Gulf of Finland. Highlights multiple interesting facts e.g. how important trade route the Baltic Sea is, how many ferries there are in the Turku archipelago and how Finns really like visiting Estonia. Each dot represents one ship. Colors represent the countries of origin.

https://tjukanov.org/gulfoffinland/

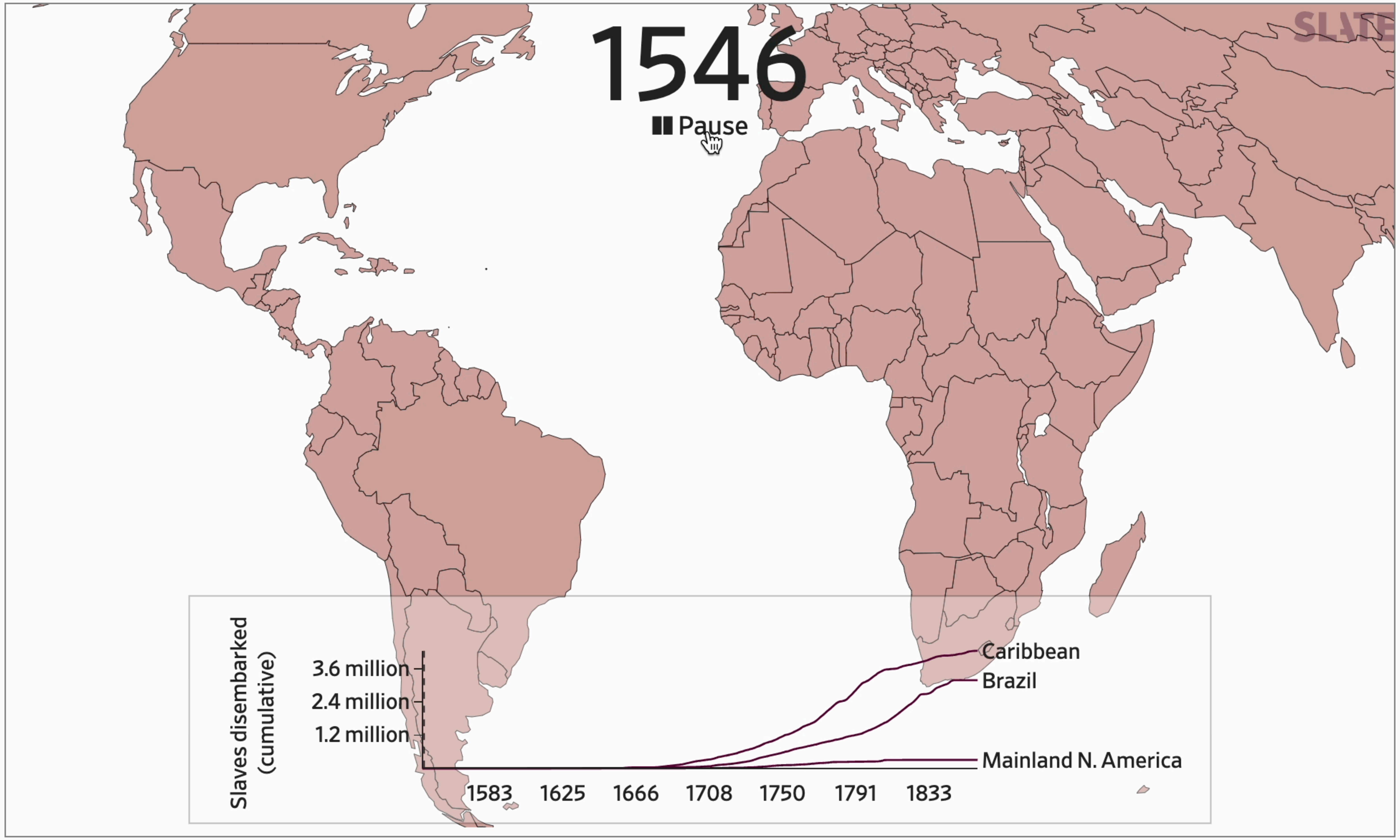


# Variable

# The Atlantic Slave Trade in Two Minutes

315 years. 20,528 voyages. Millions of lives.

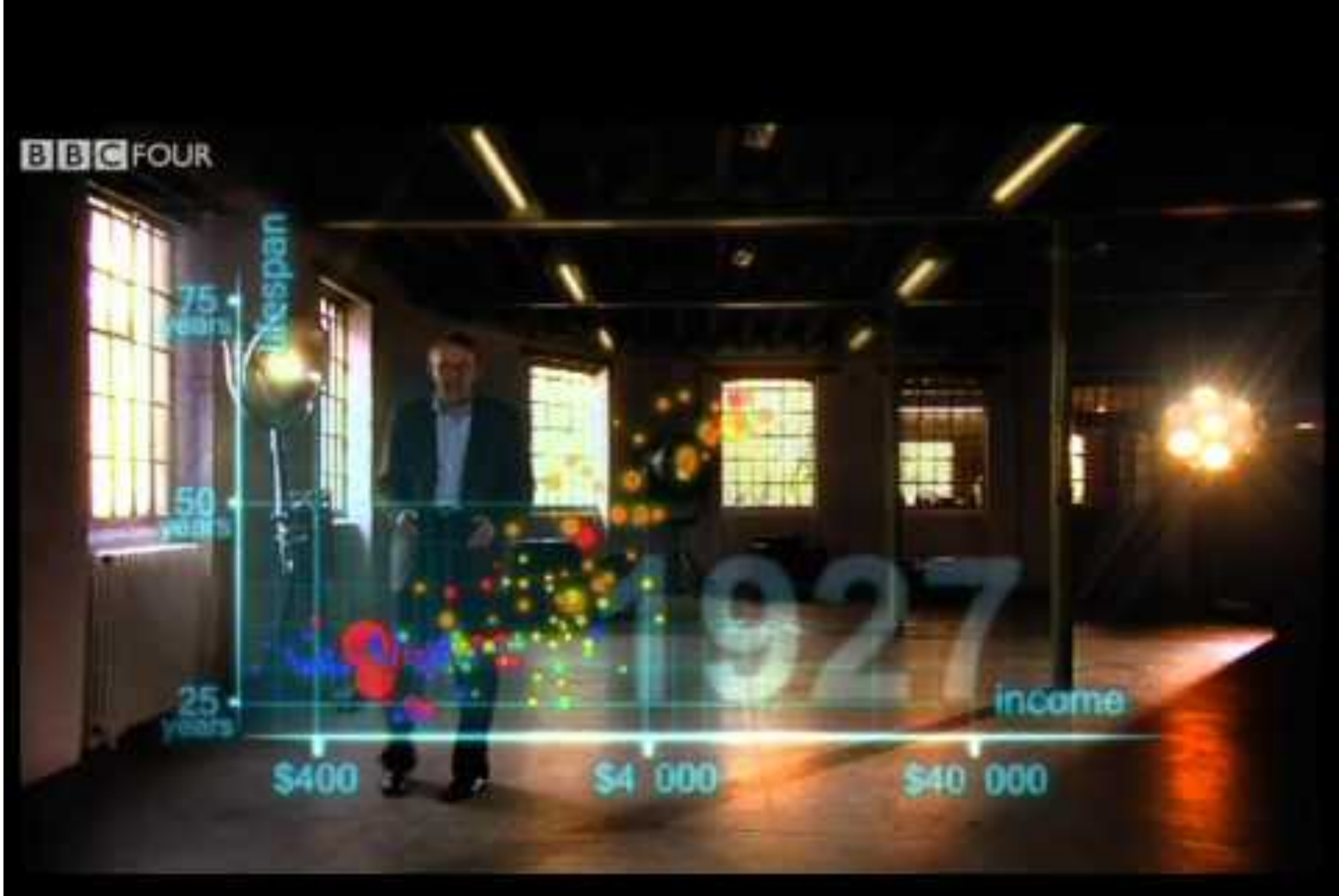
BY ANDREW KAHN AND JAMELLE BOUIE SEPT 16, 2021 • 4:18 PM




Source: slavevoyages.org

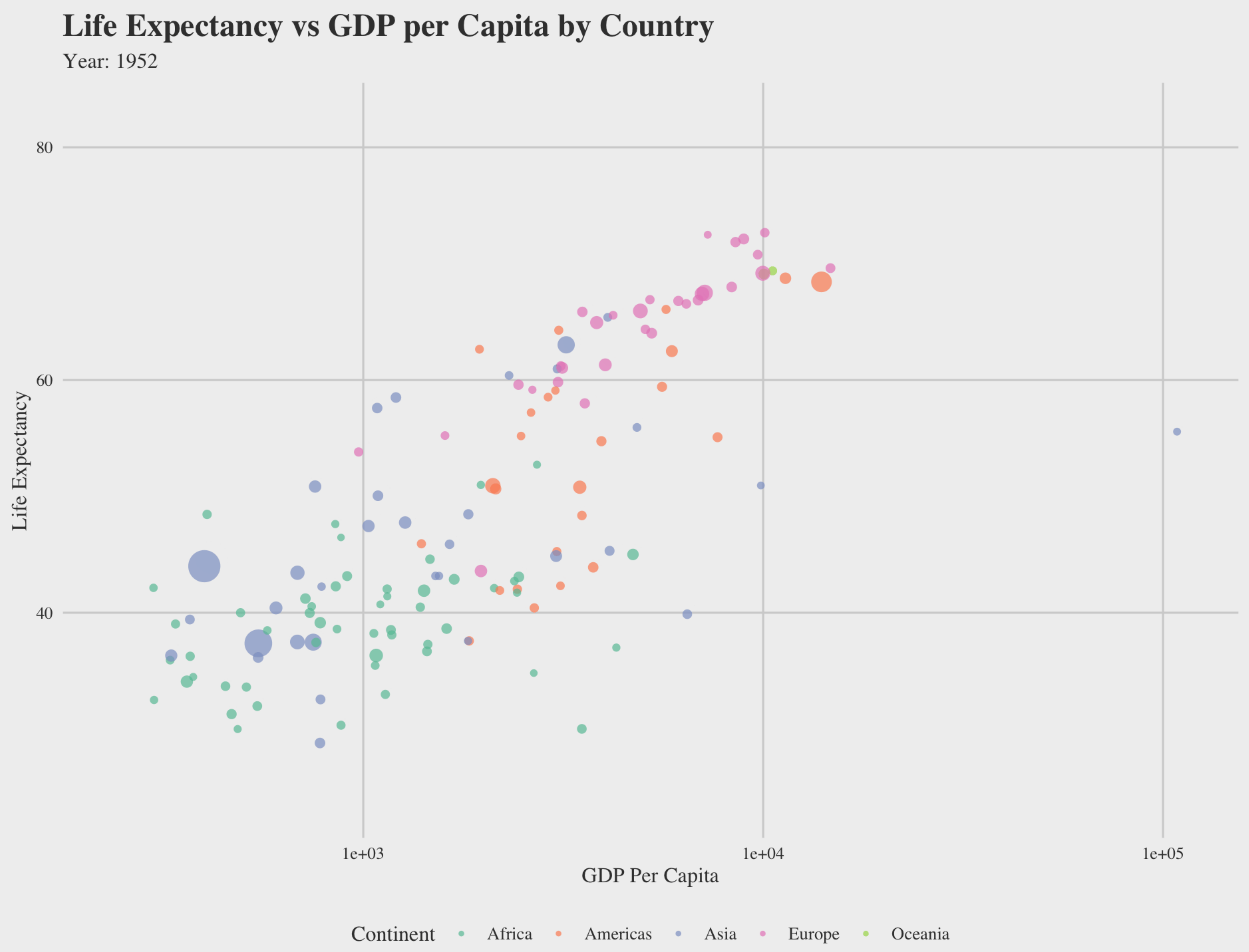
https://slate.com/news-and-politics/2021/09/atlantic-slave-trade-history-animated-interactive.html

# Variable




https://www.youtube.com/watch?v=jbkSRLYSojo&ab_channel=BBC

# vs. this




https://rpubs.com/amateiattram/codethroughproject_updated

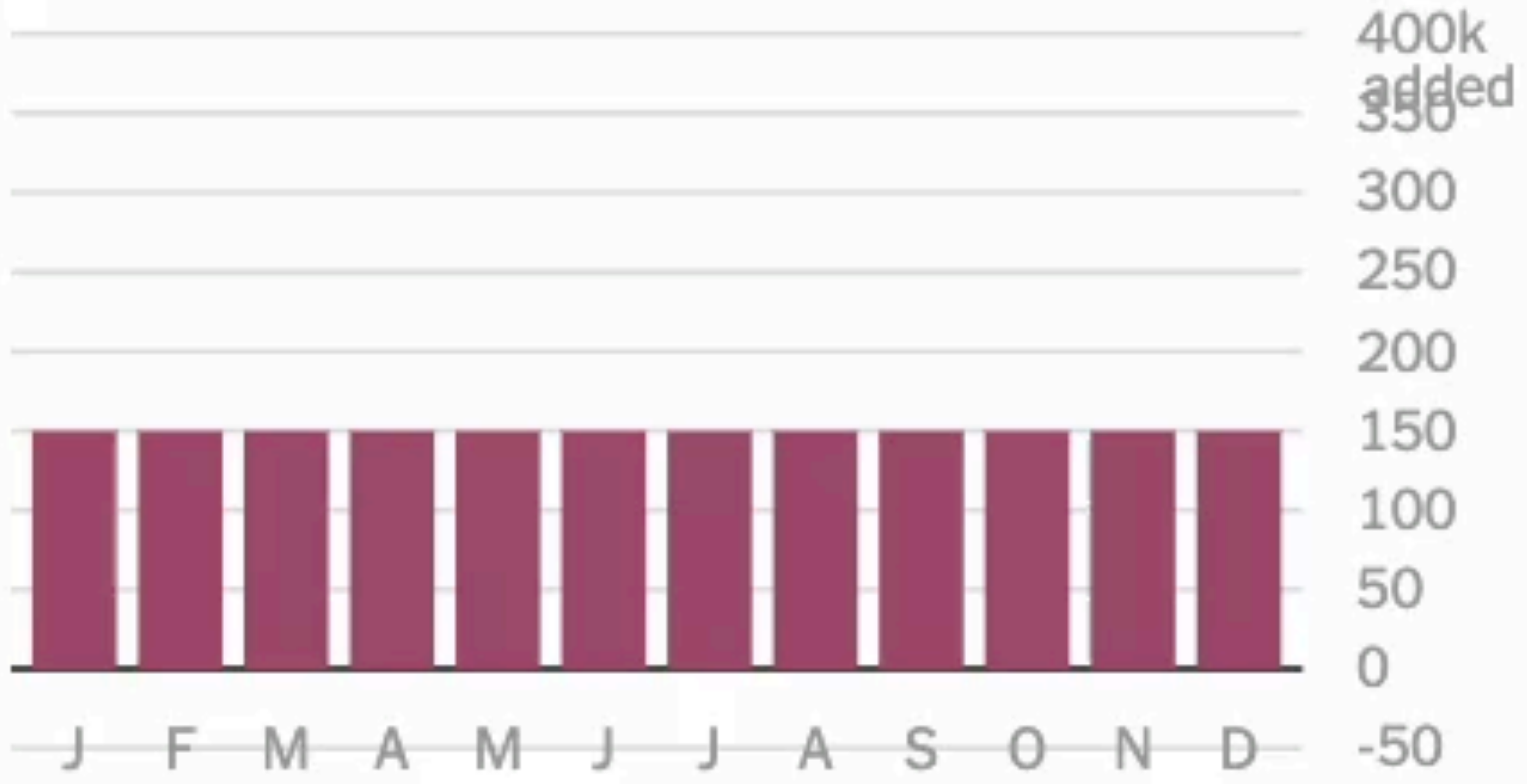
# Variable

Show uncertainty by looping over likely outcomes

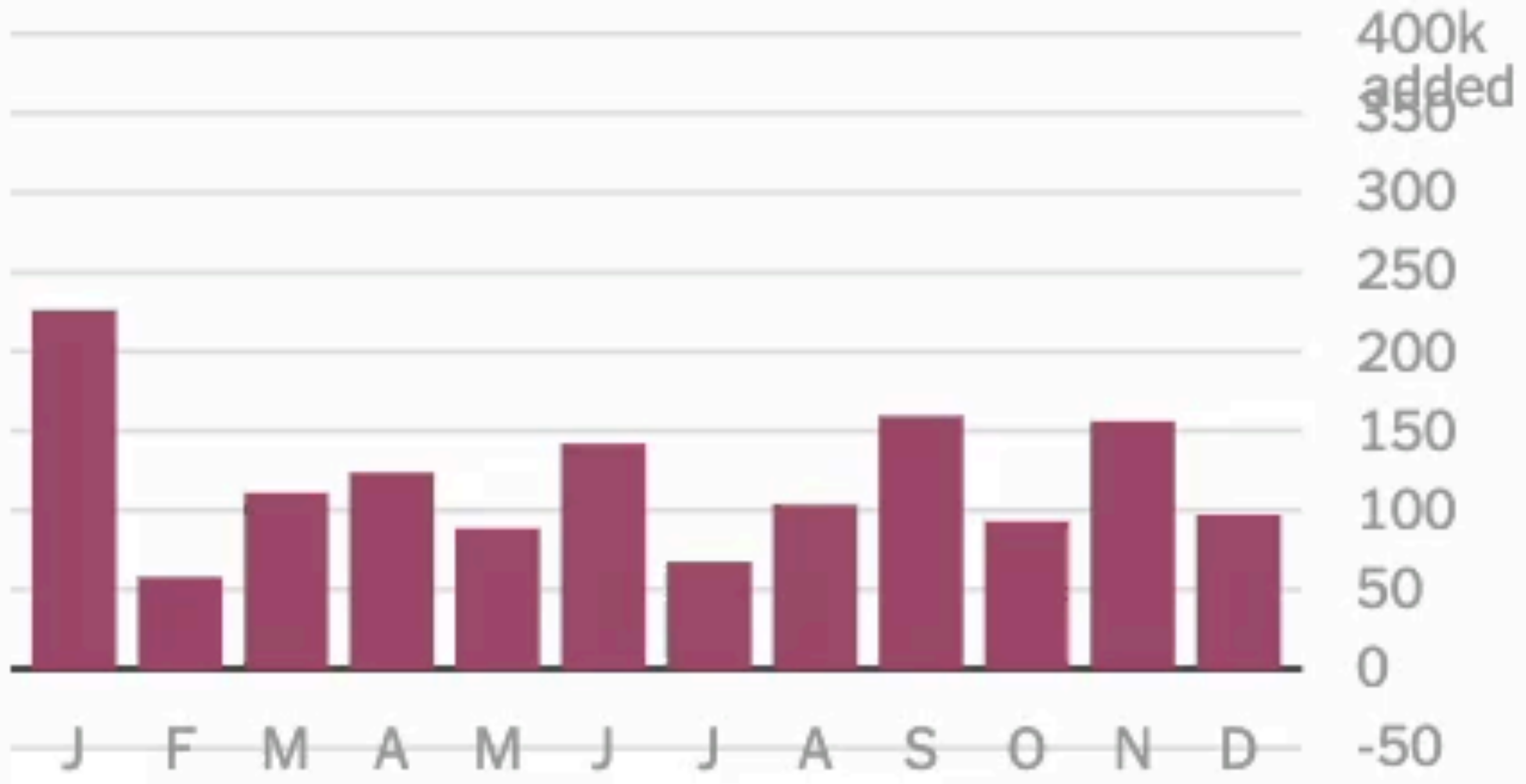
## How Not to Be Misled by the Jobs Report

Let’s consider how statistical noise might affect our perception of the jobs report in two scenarios: one with steady job growth and one with accelerating job growth.

If job growth **were actually steady** over the last 12 months...



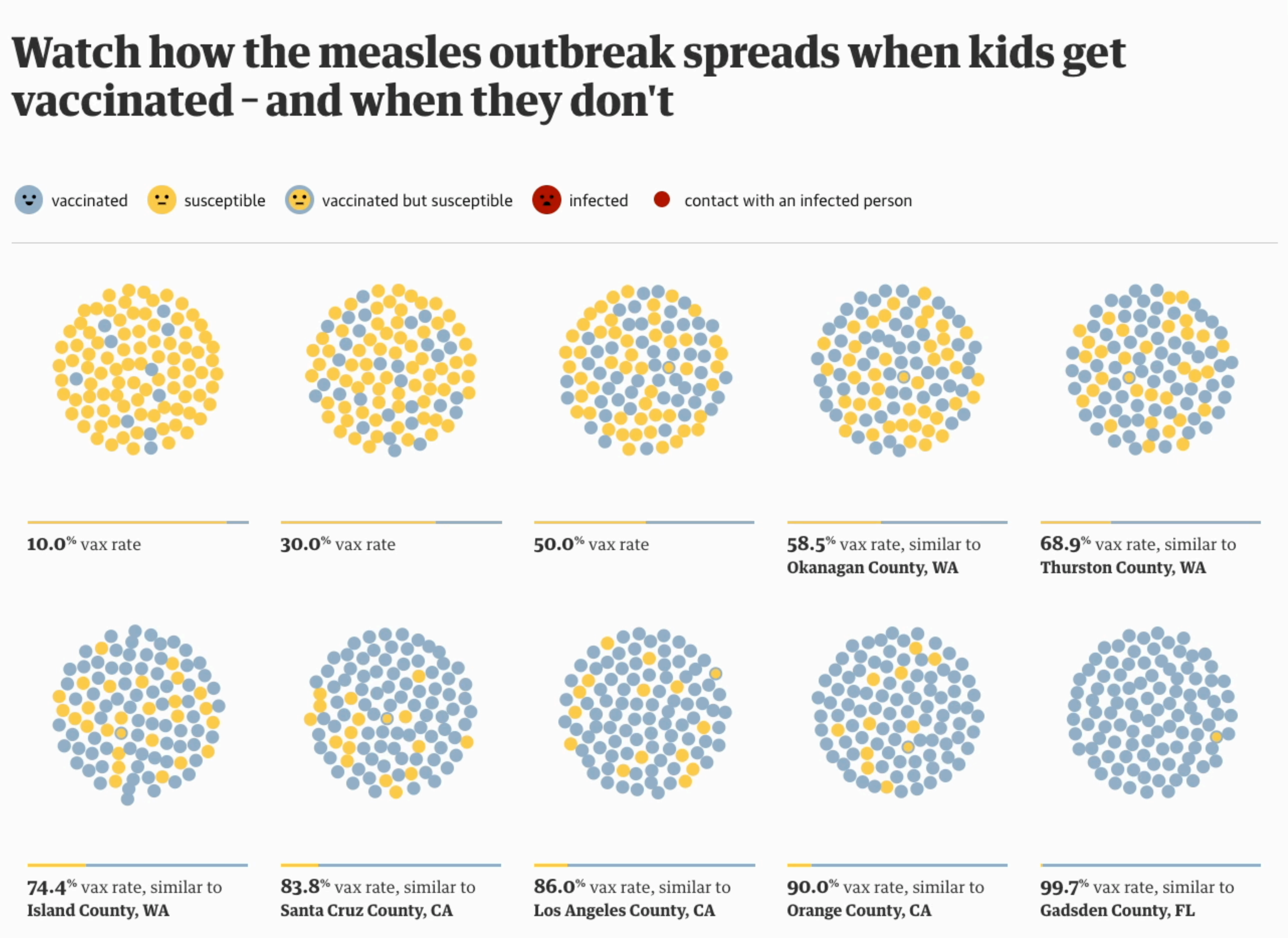

...the jobs report **could look like this**:




Did you infer a pattern from the chart on the right? If you did, you were reading too much into the jobs numbers. That chart should be a straight line, but we’ve added the same amount of sampling error that the jobs report has.

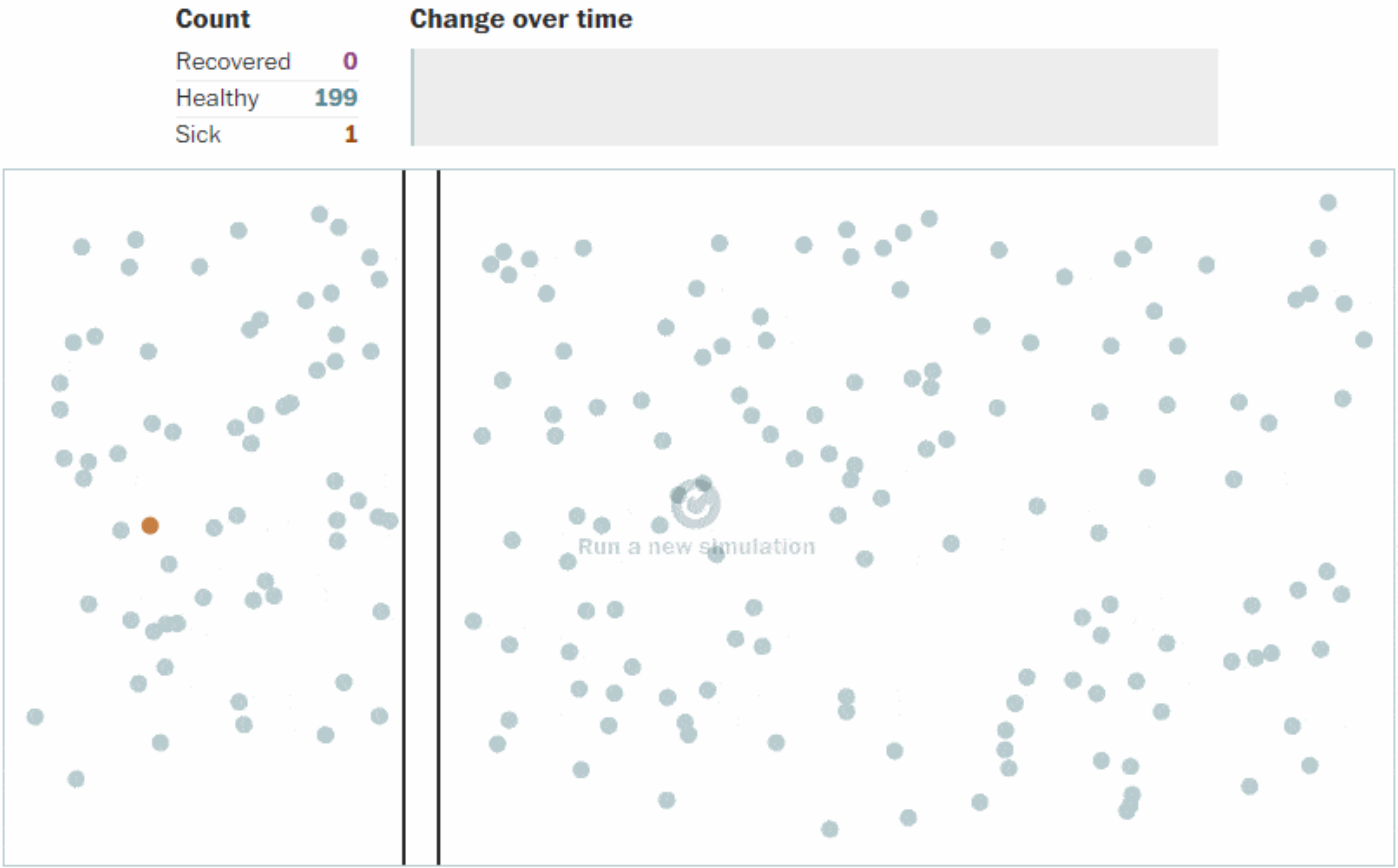
https://www.nytimes.com/2014/05/02/upshot/how-not-to-be-misled-by-the-jobs-report.html

# Variable / Process




https://www.theguardian.com/society/ng-interactive/2015/feb/05/-sp-watch-how-measles-outbreak-spreads-when-kids-get-vaccinated

# Variable / Process



# Explain a process

## The Bootstrap

Much of frequentist inference centers on the use of "good" estimators. The precise distributions of these estimators, however, can often be difficult to derive analytically. The computational technique known as the Bootstrap provides a convenient way to estimate properties of an estimator via resampling. In this example, we resample with replacement from the empirical distribution function (which is itelf generated by sampling once from the population) in order to estimate the standard error of the sample mean.

Choose a probability distribution from which we will sample once to generate the empirical distribution function.

-- select a distribution --

Choose a sample (and resampling) size ($n$) and sample from your chosen distribution.

$n = 10$ Sample

Resample to get an idea of the spread of the sample mean's distribution.

Resample | Resample 100 times

distribution

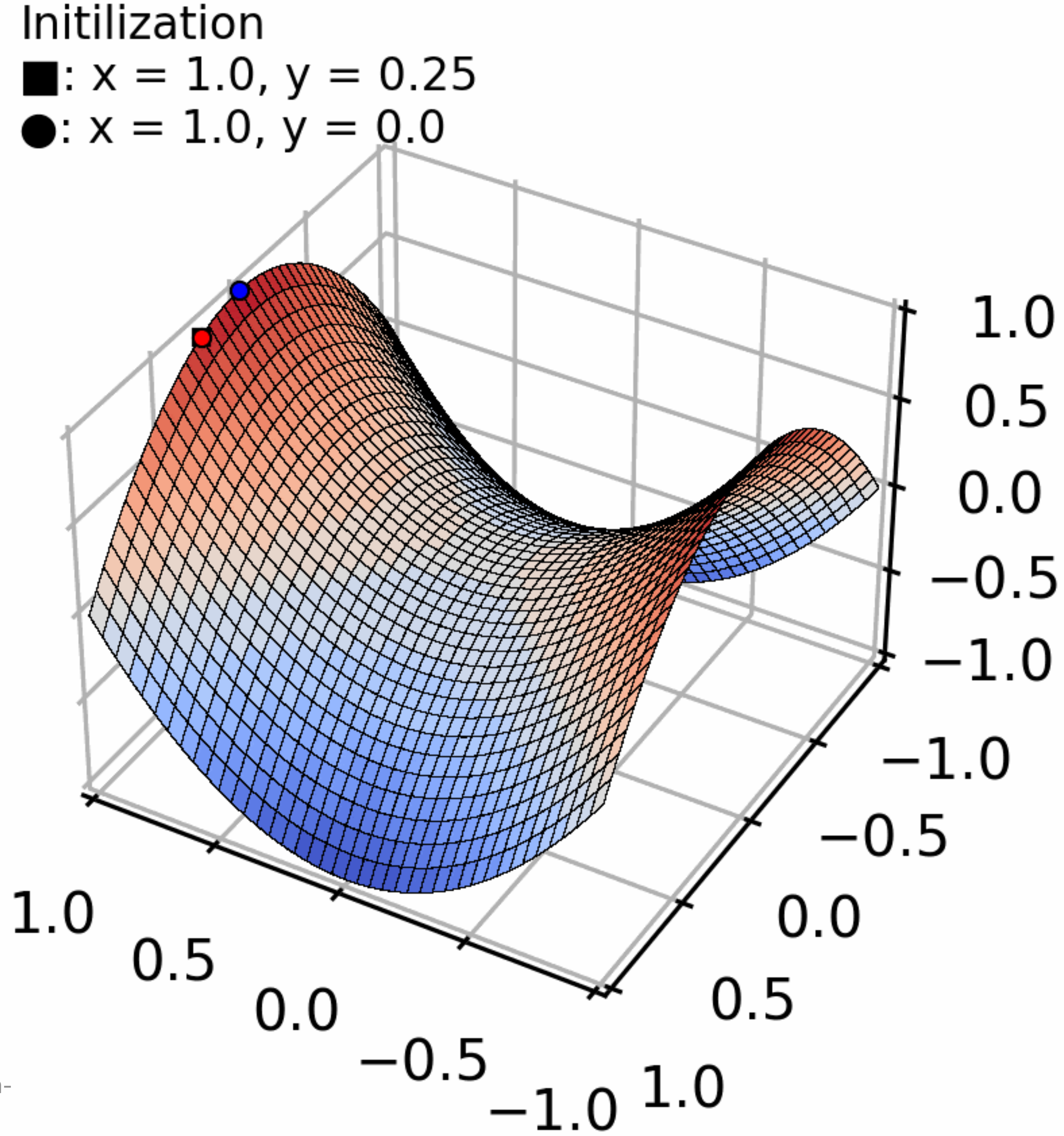
sample

resample + average

count

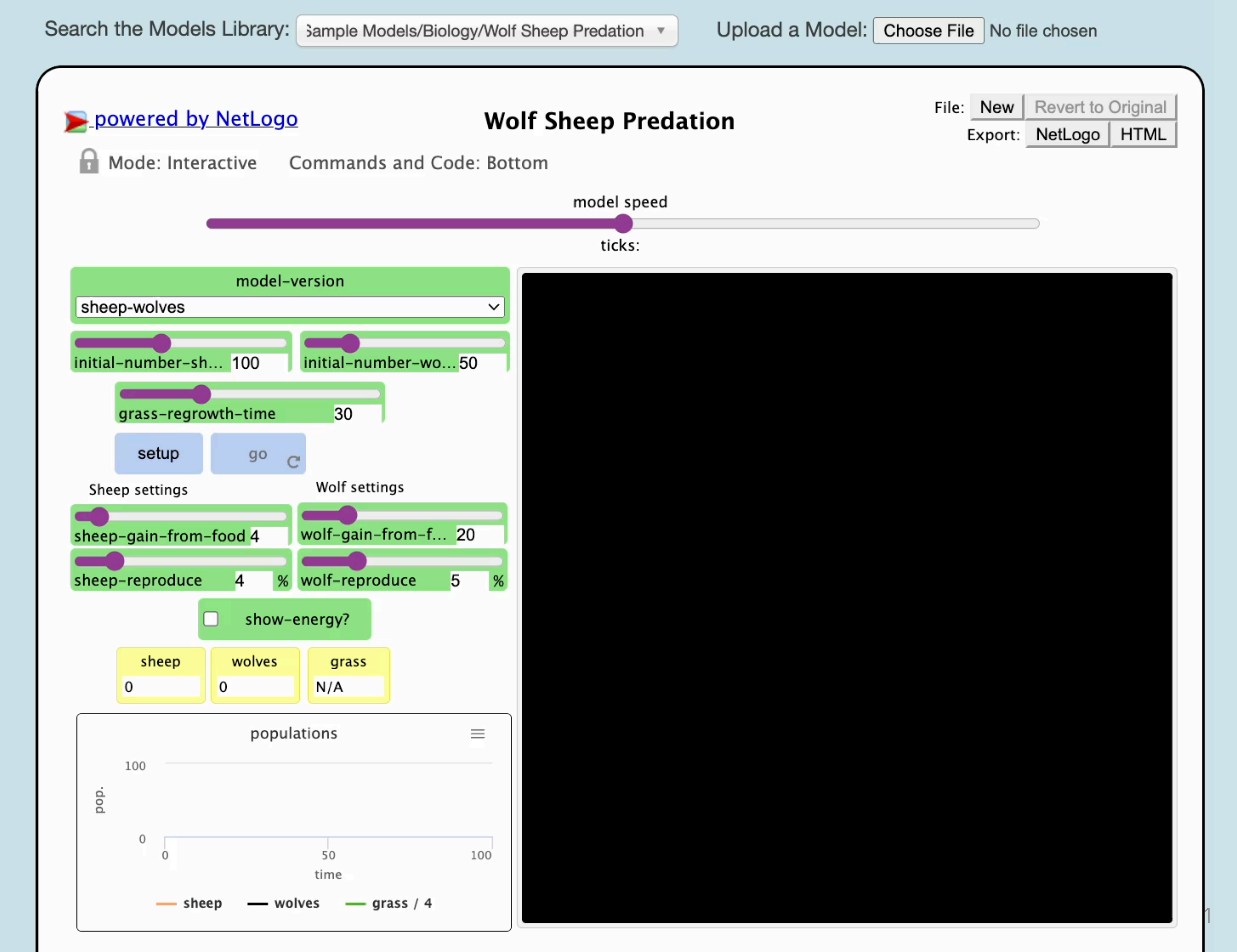
https://seeing-theory.brown.edu/frequentist-inference/index.html#section3

# Explain a process




https://towardsdatascience.com/creating-a-gradient-descent-animation-in-python-3c4dcd20ca51-2/

# Explain a process




https://www.netlogoweb.org/launch#http://www.netlogoweb.org/assets/modelslib/Sample%20Models/Biology/Wolf%20Sheep%20Predation.nlogo

# Apple's Value Hit $1 Trillion. Add Disney to Bank of America and … You're Halfway There

By JON HUANG, KARL RUSSELL and JACK NICAS AUG. 2, 2018

START

https://www.nytimes.com/interactive/2018/08/02/technology/apple-trillion-market-cap.html

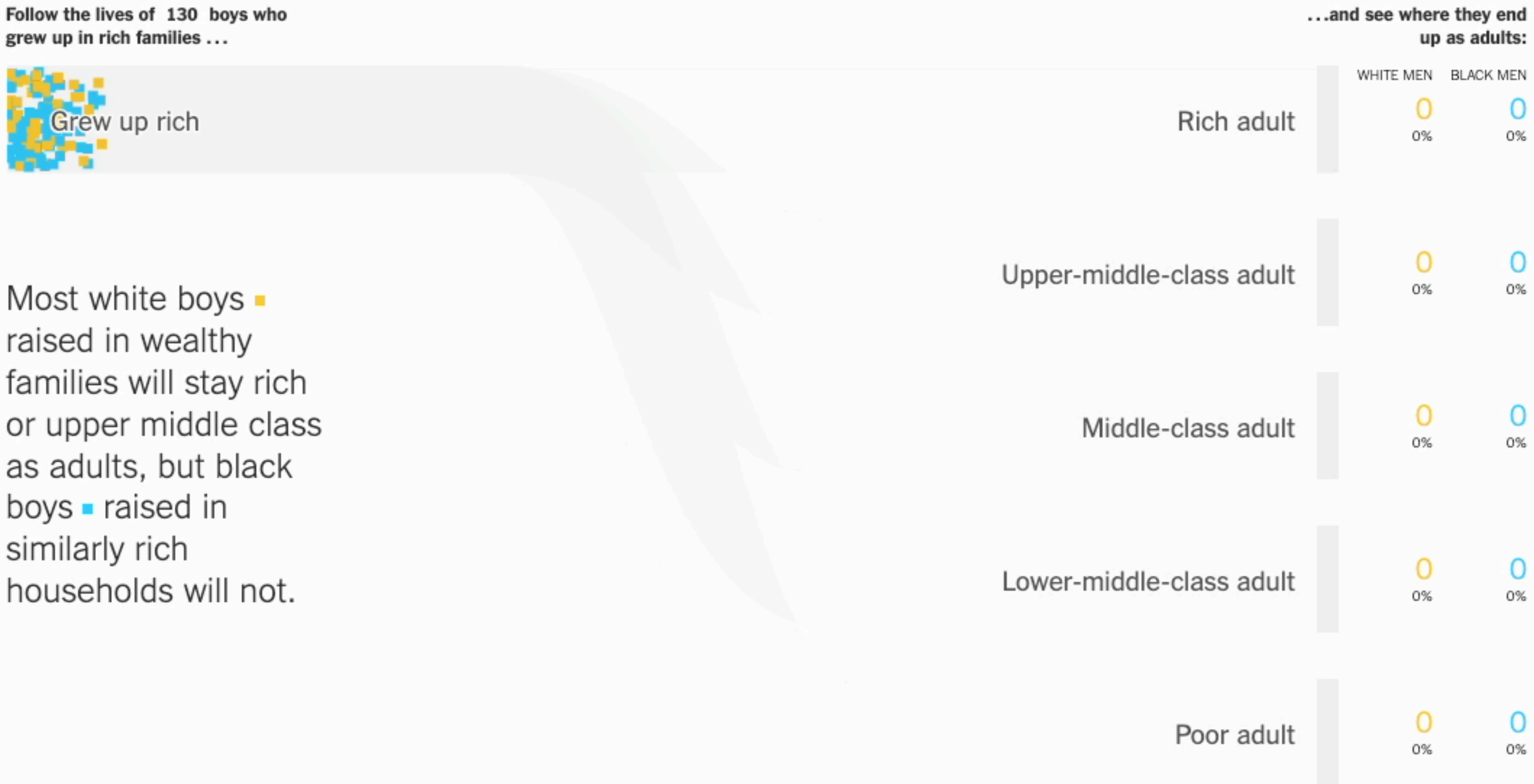
# Engagement

## *Extensive Data Shows Punishing Reach of Racism for Black Boys*

By **EMILY BADGER, CLAIRE CAIN MILLER, ADAM PEARCE** and **KEVIN QUEALY** MARCH 19, 2018

Black boys raised in America, even in the wealthiest families and living in some of the most well-to-do neighborhoods, still earn less in adulthood than white boys with similar backgrounds, according to a sweeping new study that traced the lives of millions of children.

White boys who grow up rich are likely to remain that way. Black boys raised at the top, however, are more likely to become poor than to stay wealthy in their own adult households.



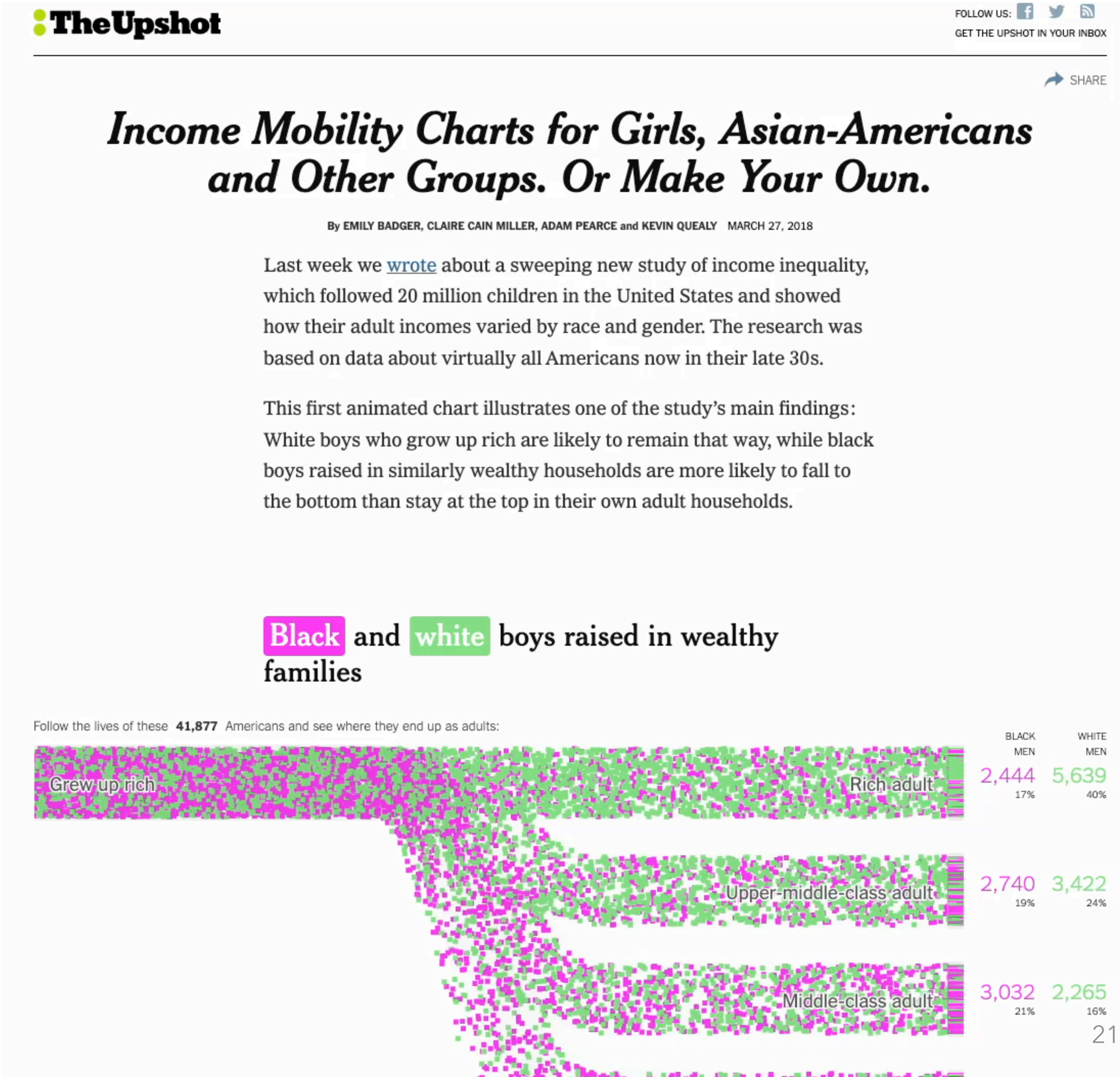

https://www.nytimes.com/interactive/2018/03/19/upshot/race-class-white-and-black-men.html

# Group activity

## Scroll all way down

https://www.nytimes.com/interactive/2018/03/27/upshot/make-your-own-mobility-animation.html

## Select a combination, and share your thoughts


TheUpshot

FOLLOW US:

GET THE UPSHOT IN YOUR INBOX

SHARE

**_Income Mobility Charts for Girls, Asian-Americans and Other Groups. Or Make Your Own._**

By EMILY BADGER, CLAIRE CAIN MILLER, ADAM PEARCE and KEVIN QUEALY MARCH 27, 2018

Last week we wrote about a sweeping new study of income inequality, which followed 20 million children in the United States and showed how their adult incomes varied by race and gender. The research was based on data about virtually all Americans now in their late 30s.

This first animated chart illustrates one of the study's main findings: White boys who grow up rich are likely to remain that way, while black boys raised in similarly wealthy households are more likely to fall to the bottom than stay at the top in their own adult households.

Black and white boys raised in wealthy families

Follow the lives of these **41,877** Americans and see where they end up as adults:

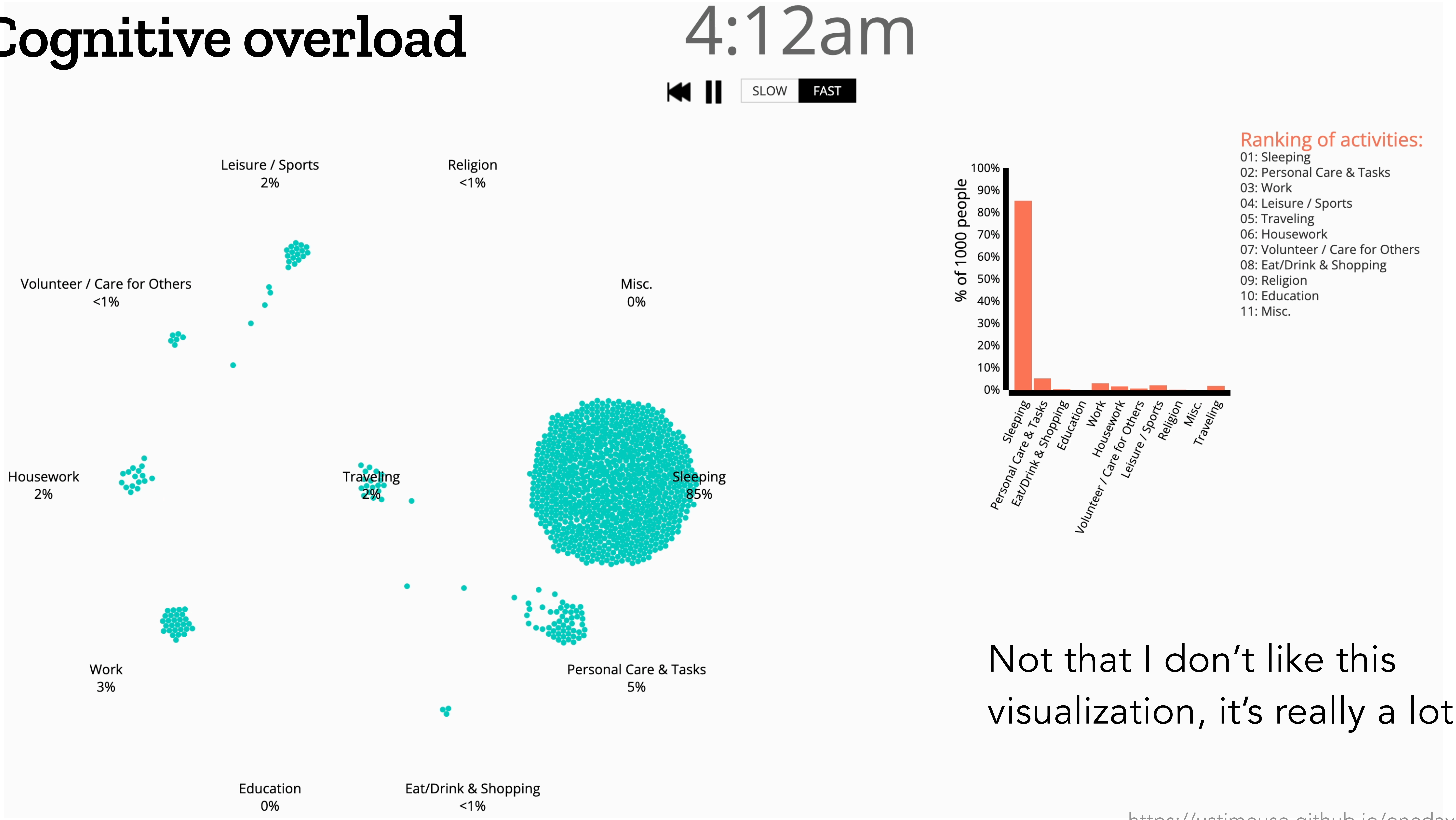
# Animations we struggle with

# Distracting

Animations can be distracting

Were you listening to me?

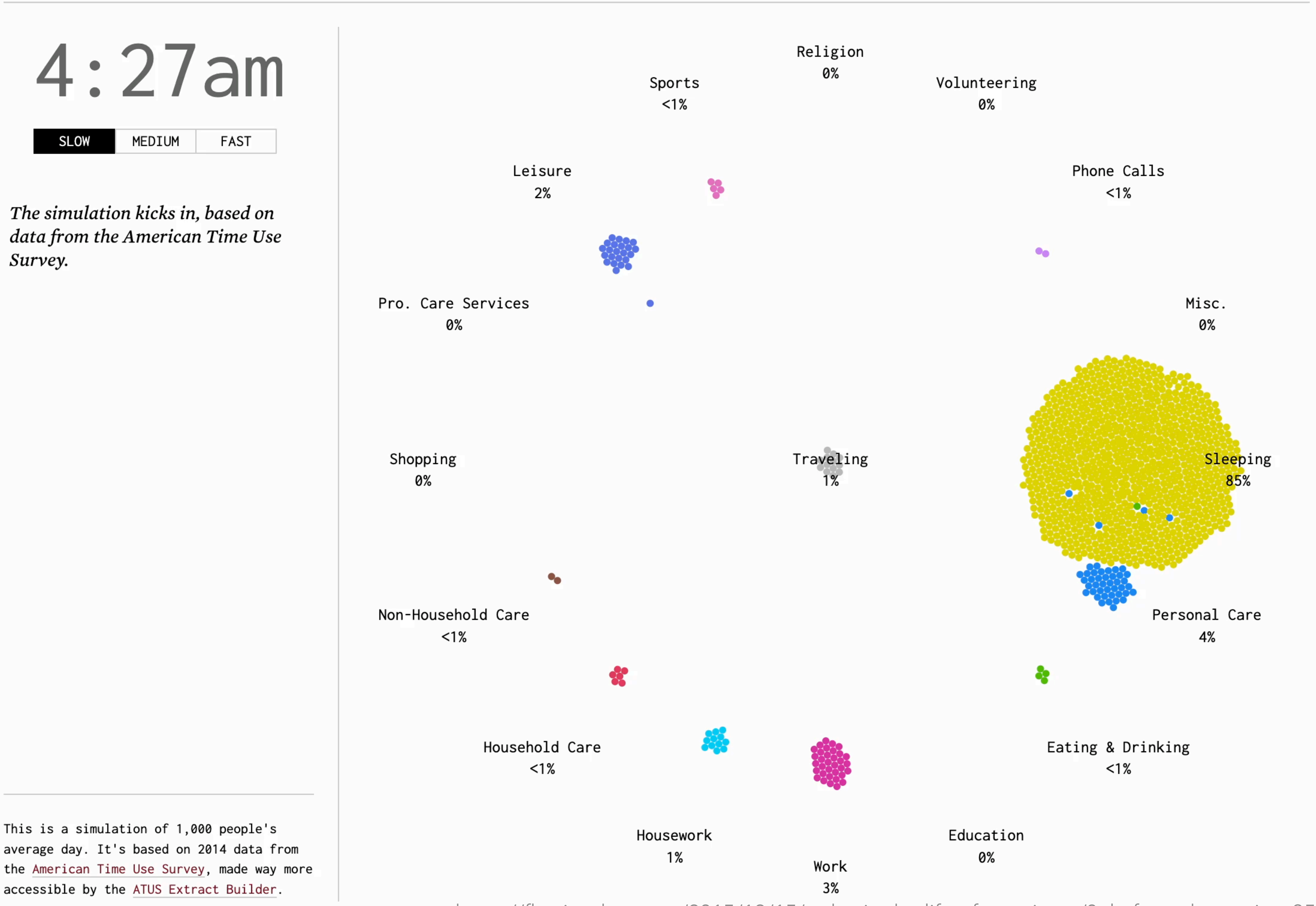
# Cognitive overload




Not that I don't like this visualization, it's really a lot

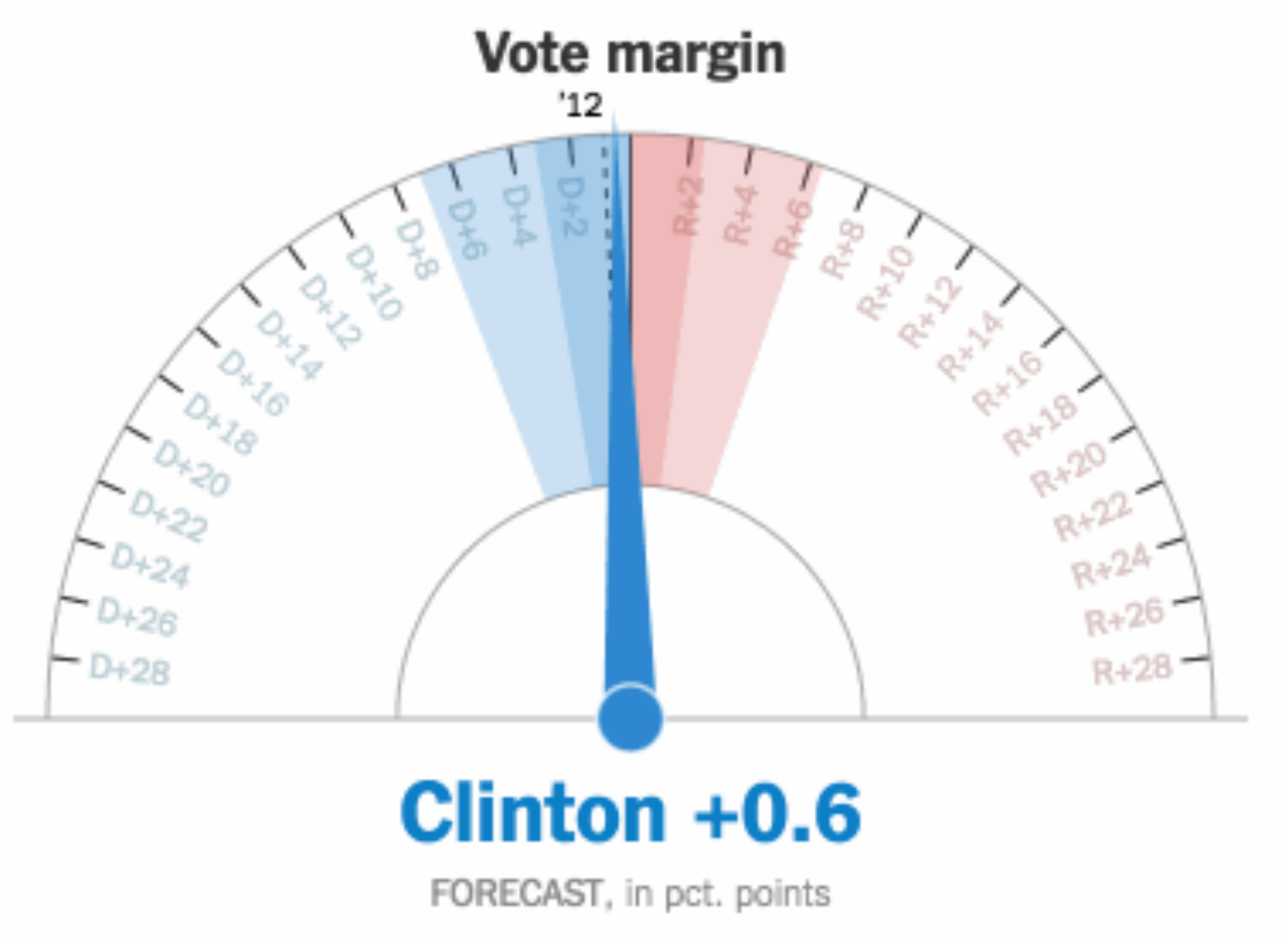
https://ustimeuse.github.io/oneday.html

# Tedious

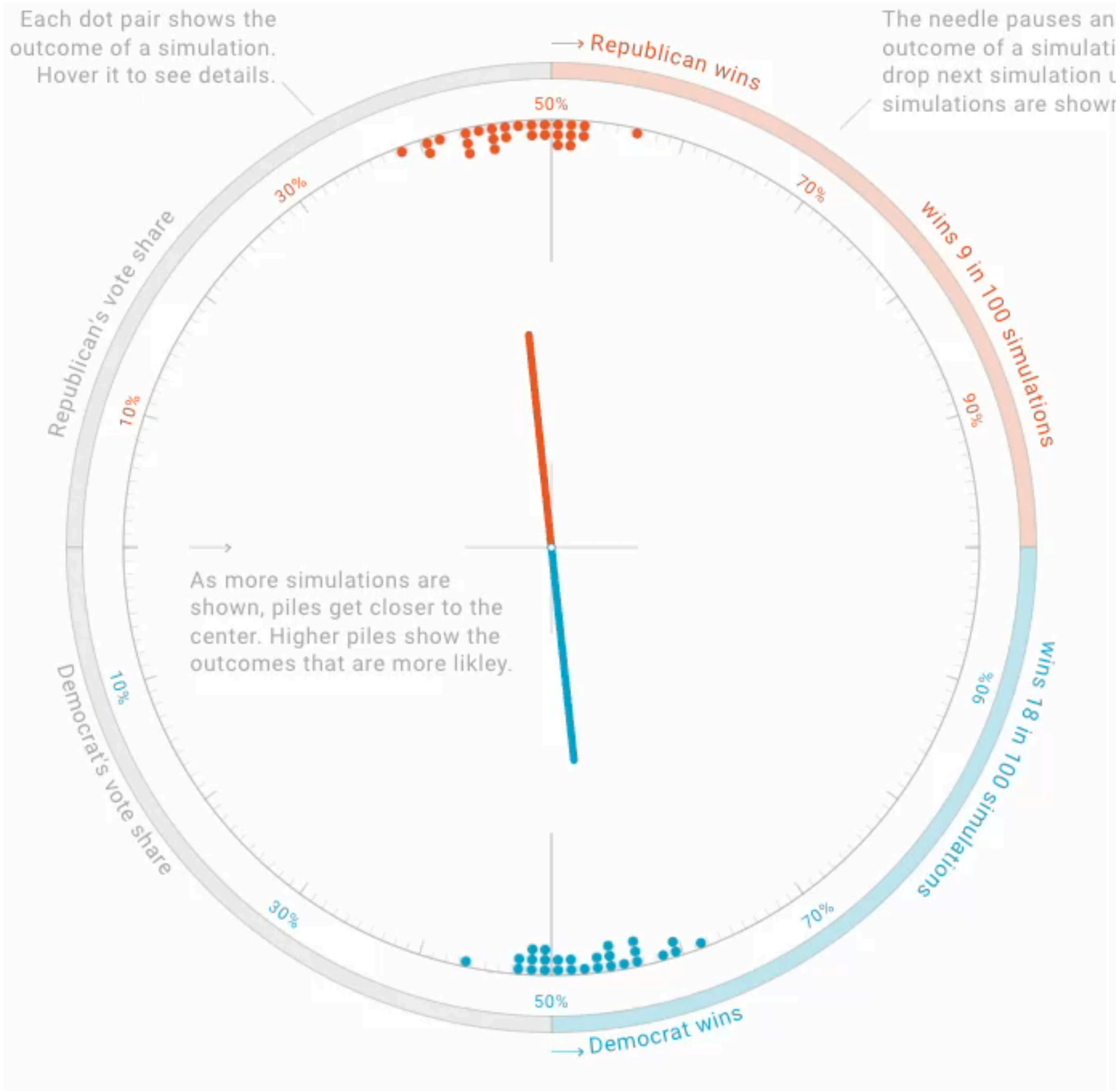



https://flowingdata.com/2015/12/15/a-day-in-the-life-of-americans/?platform=hootsuite

# Confusing




Unsure why the needle is moving….

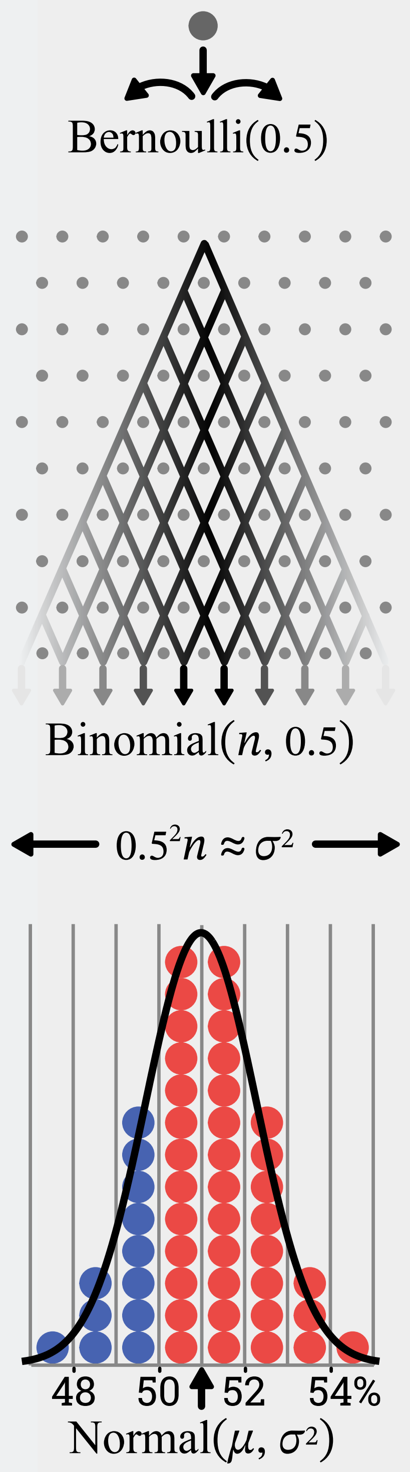


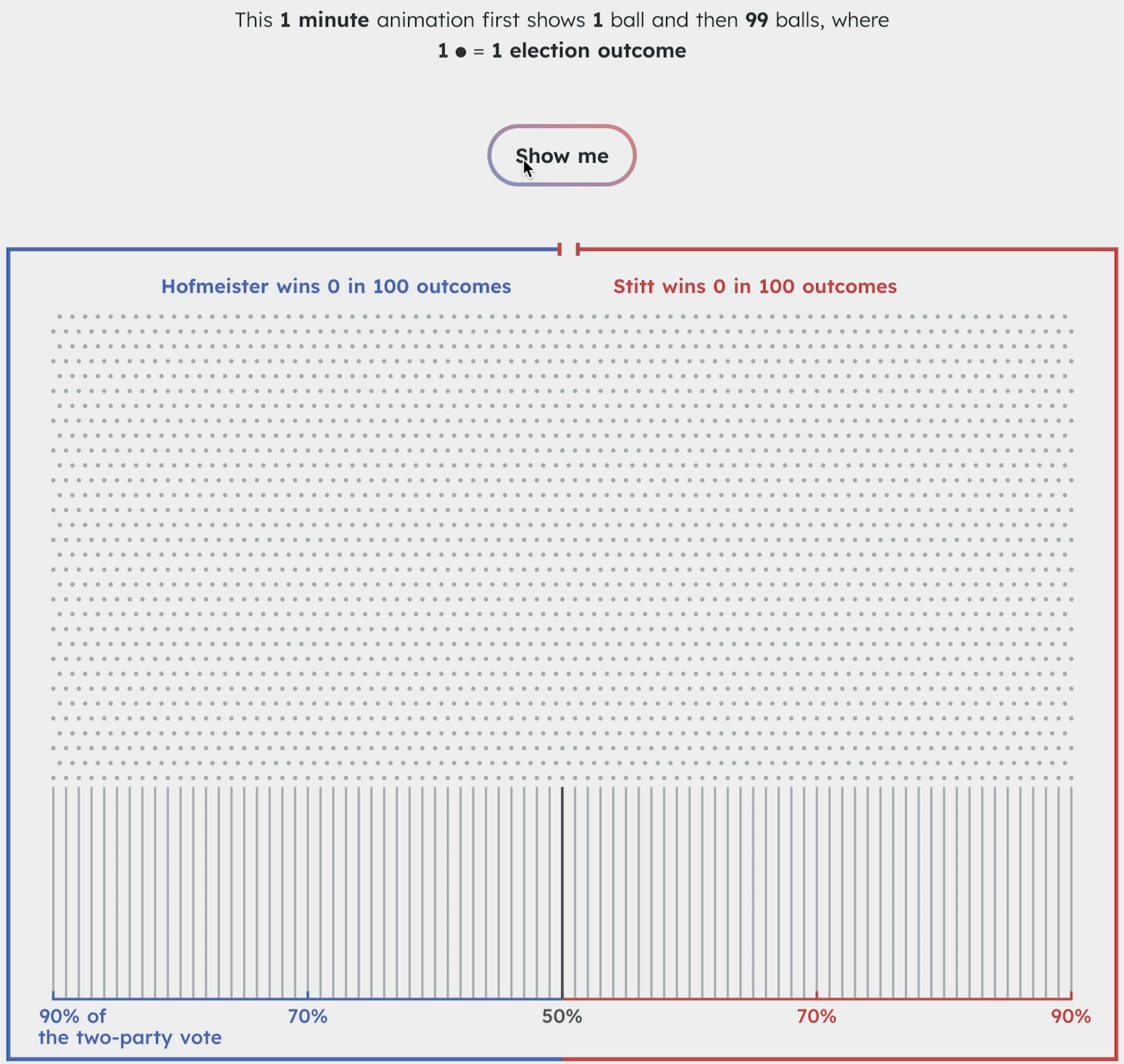

https://flourish.studio/blog/animated-gauge-visualization/

# Unscientific

We use a Plinko broad to show a stats process to avoid confusion.

People: "it's unscientific."



# Unscientific


https://www.youtube.com/watch?v=-gphdkGL4uE&t=146s

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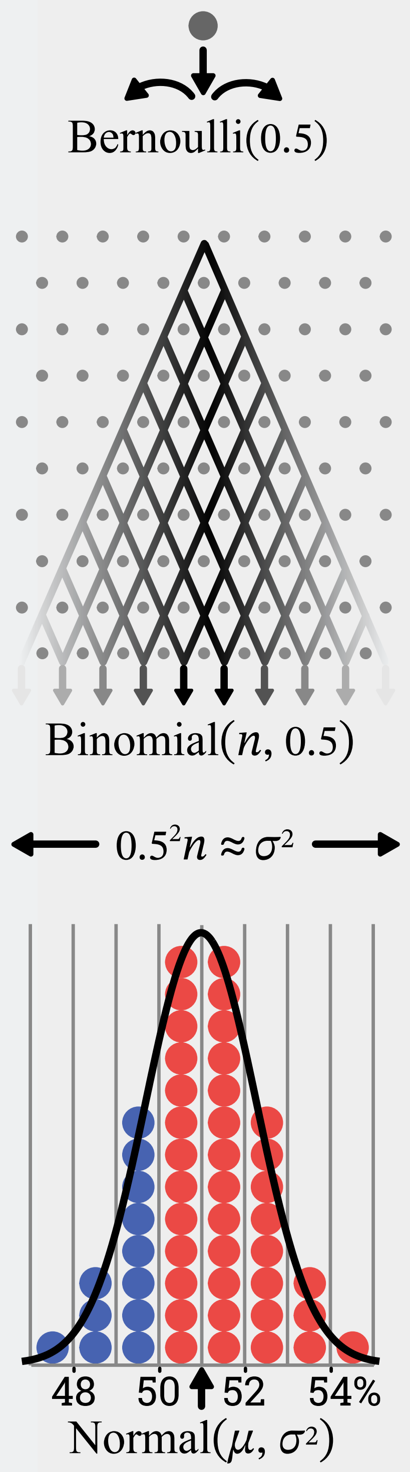
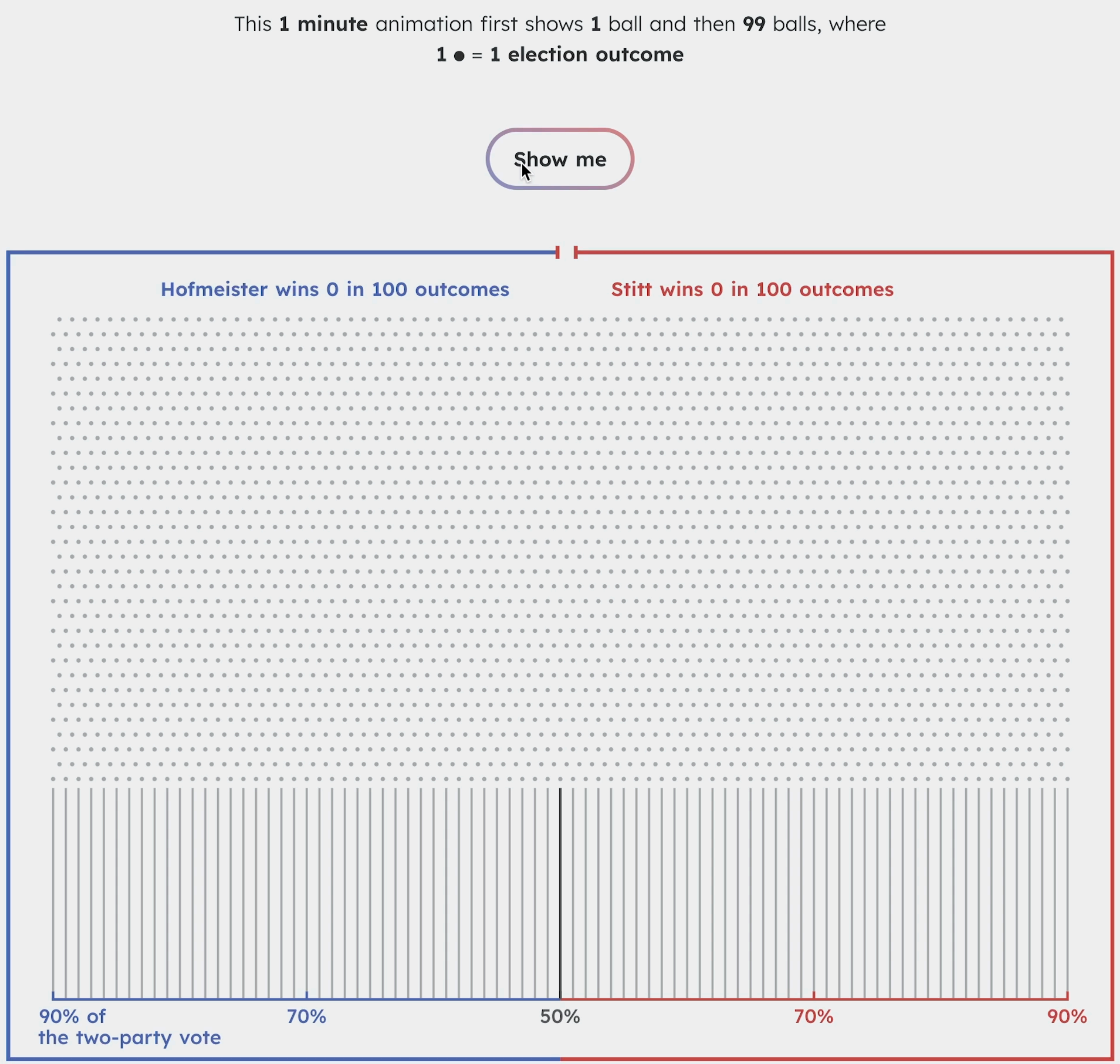


# Implementation and design

# Implementation

## Frame-Based Animation

Redraw the scene at regular intervals (e.g., 16ms).

Developer defines the redraw function (e.g., Processing, p5.js)

## Transition-Based Animation

Specify a property value, duration, and an "easing" function.

Also called **tweening** (for "in-betweens").

Steps computed via **interpolation**

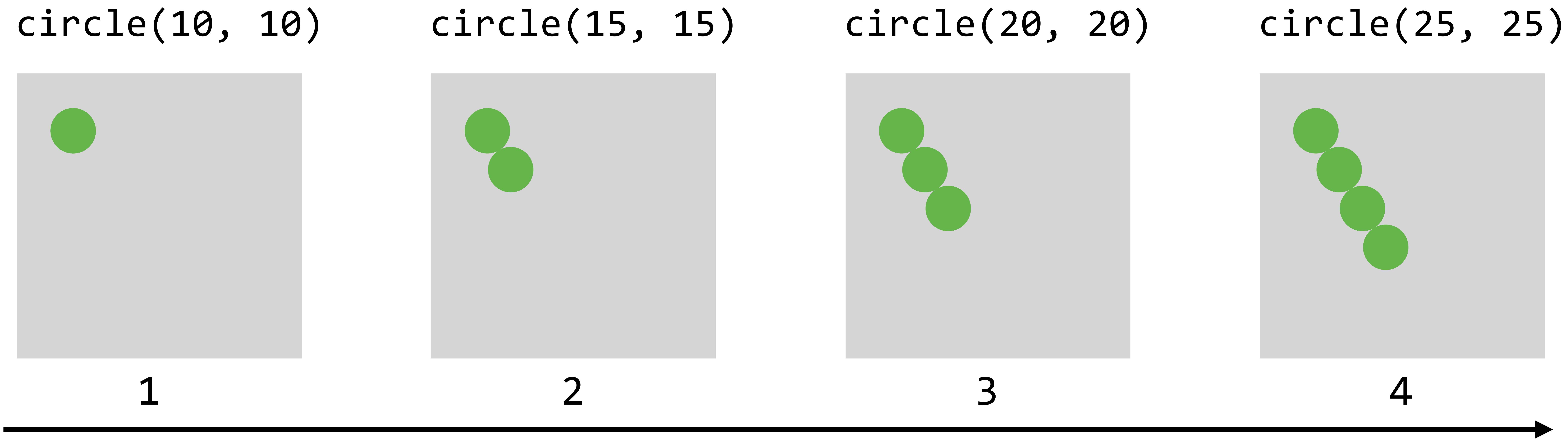
```
step (fraction) { val_now = val_start + fraction * (val_end - val_start); }
```

Timing & redraw managed by UI toolkit.

# Frame-based animation

Redraw the scene at regular intervals (e.g., 16ms).

Developer defines the redraw function (e.g., Processing, p5.js).



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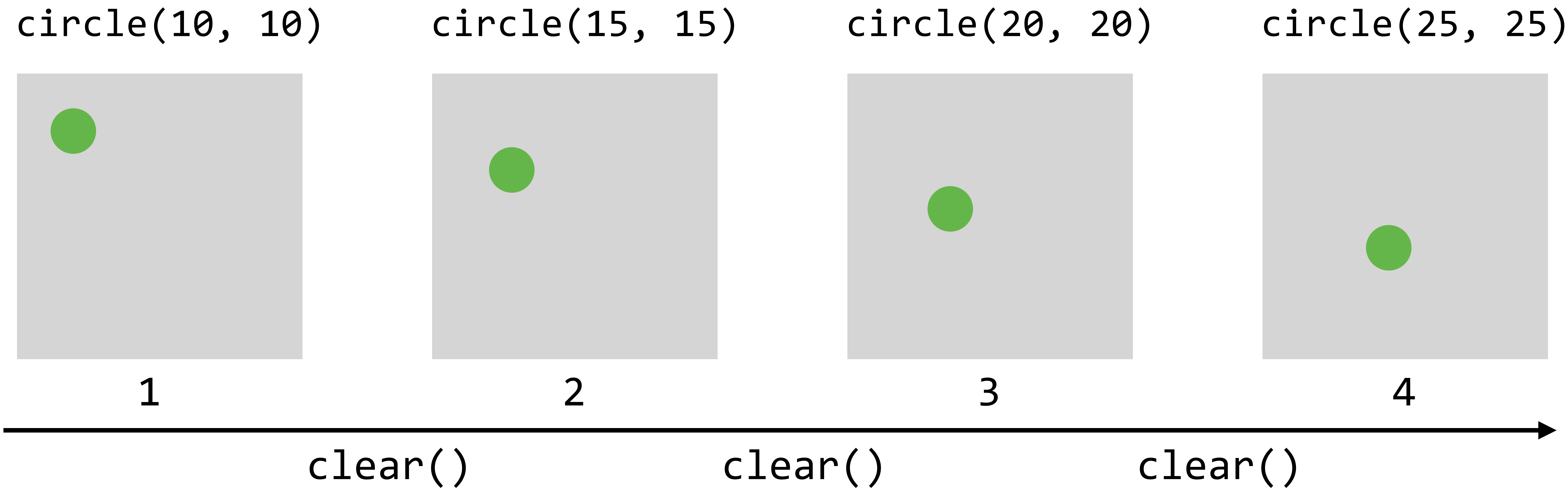
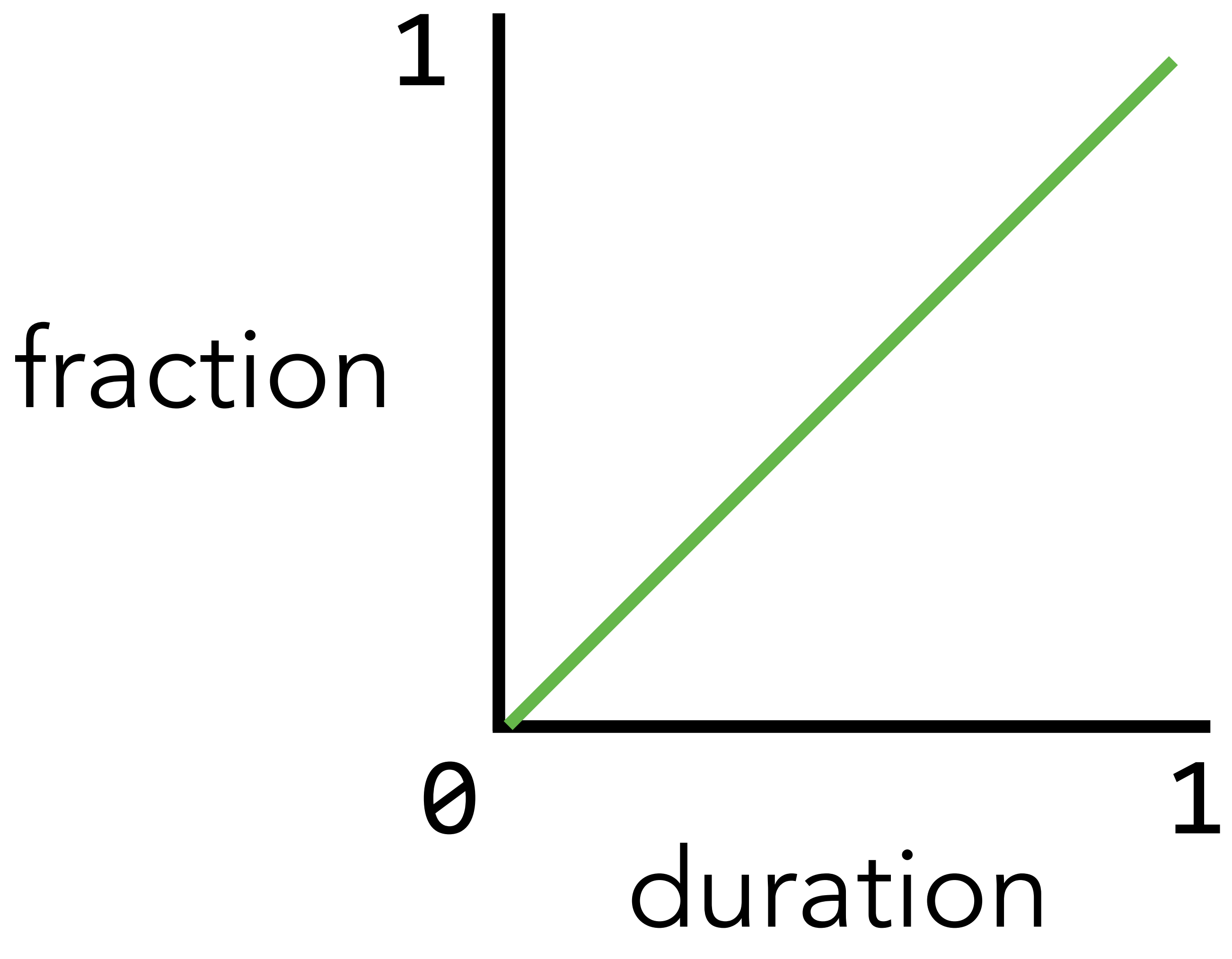


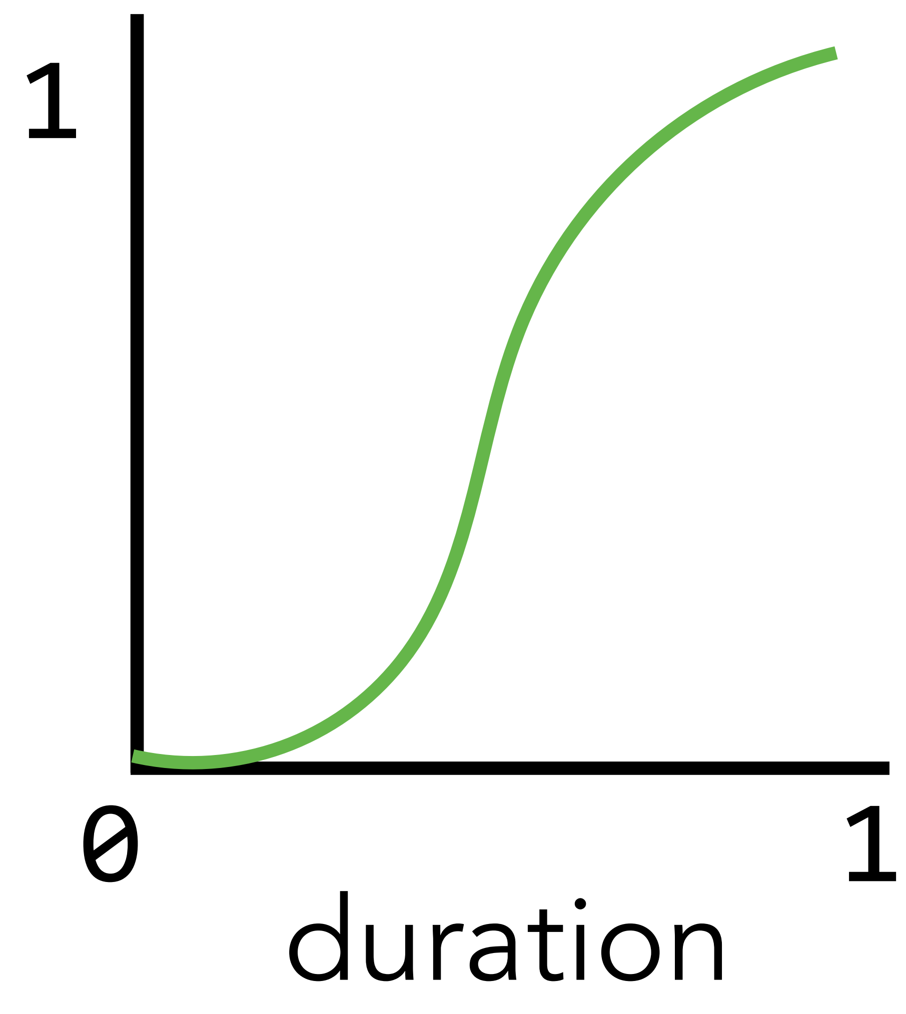
# Easing/pacing functions

**Goals:** Stylize animation, improve perception.

The idea is to **warp time**: as *duration* goes from start (0%) to end (100%), dynamically adjust the *interpolation fraction* using an easing function.



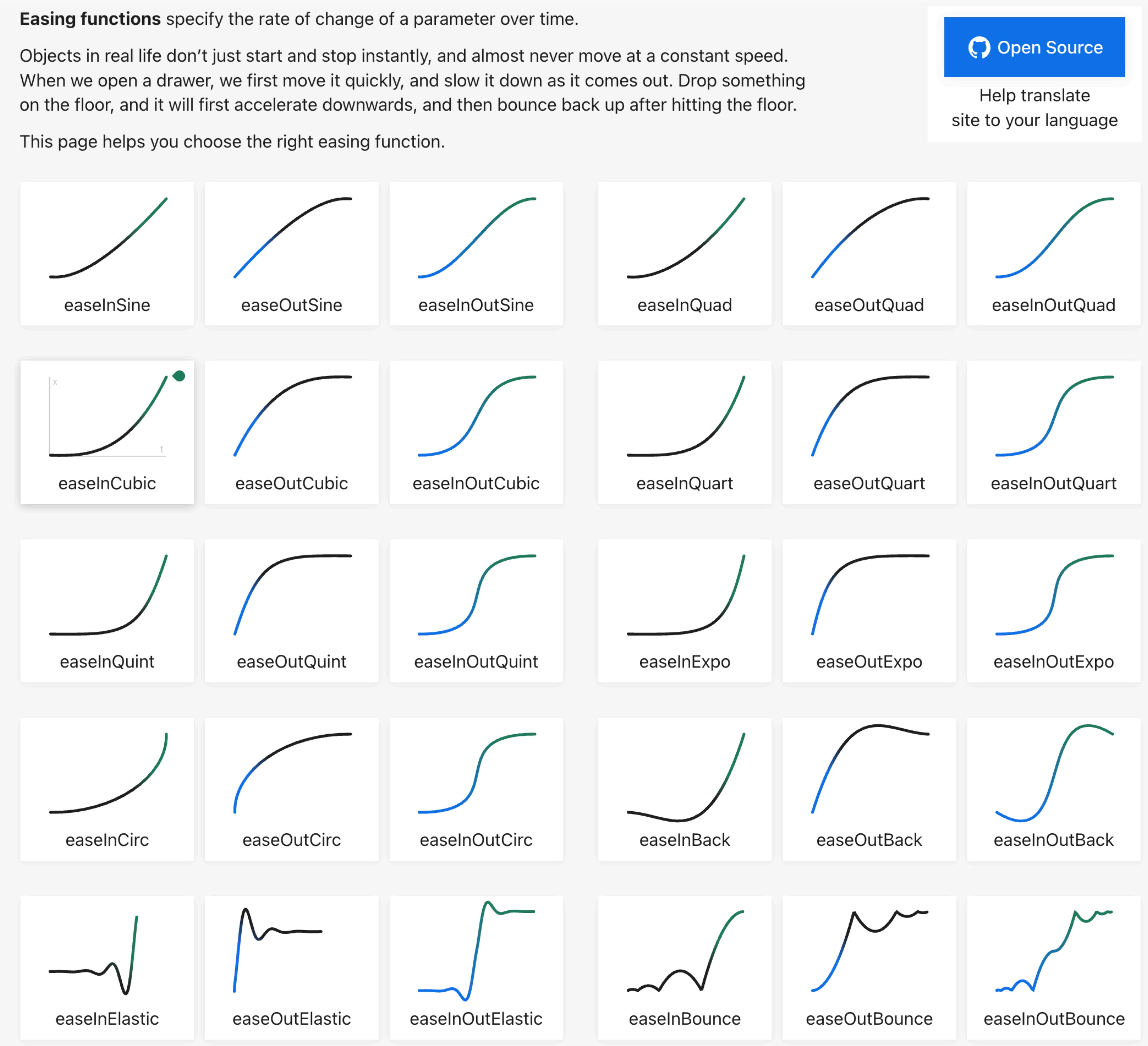

`ease(x) = x`
`(linear, no warp)`




`ease(x) = s-curve(x)`
`(slow-in, slow-out)`

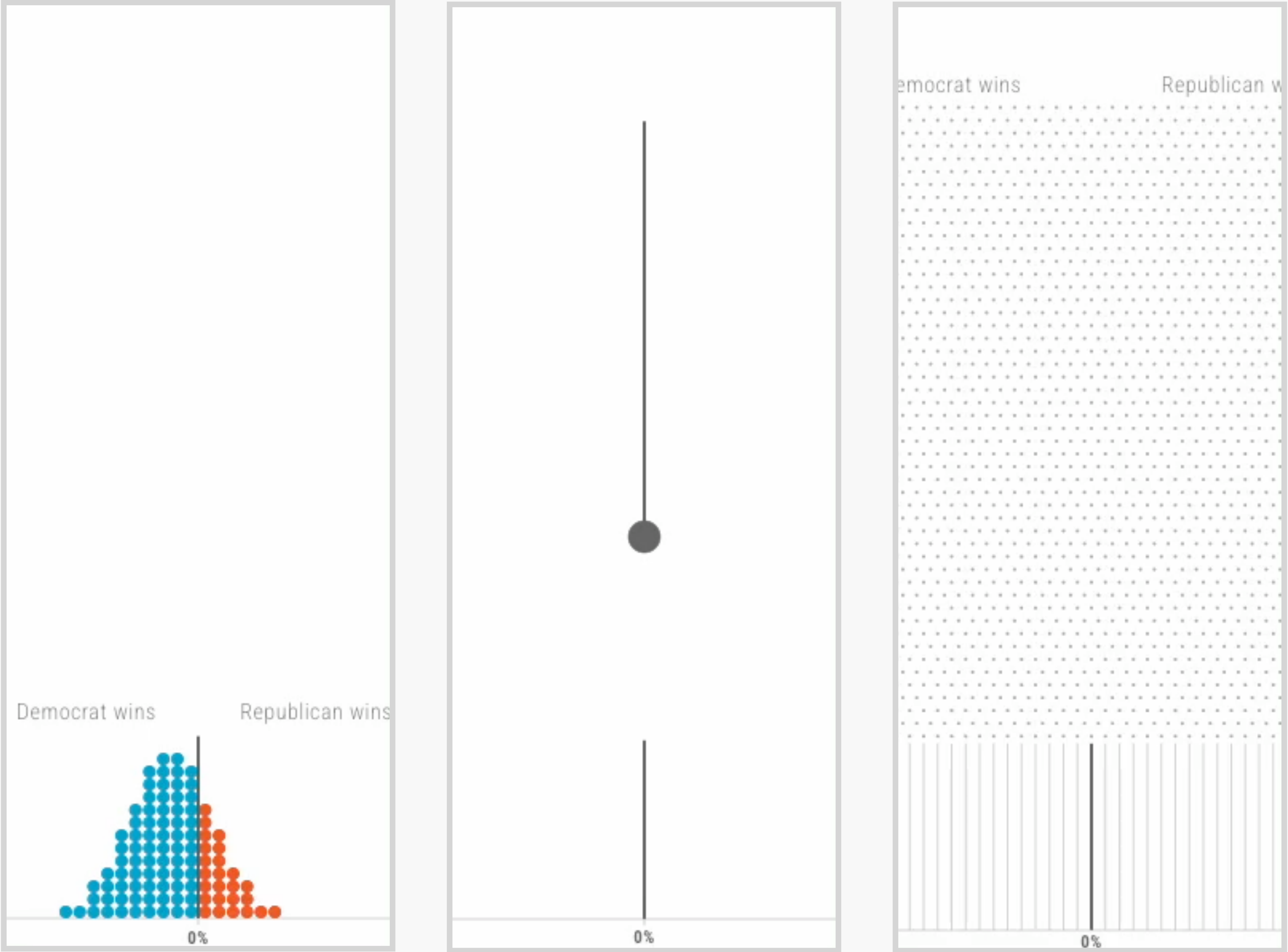
# Easing/pacing

https://easings.net/en



# Analogy

Having a good analogy seems to resolve some confusion



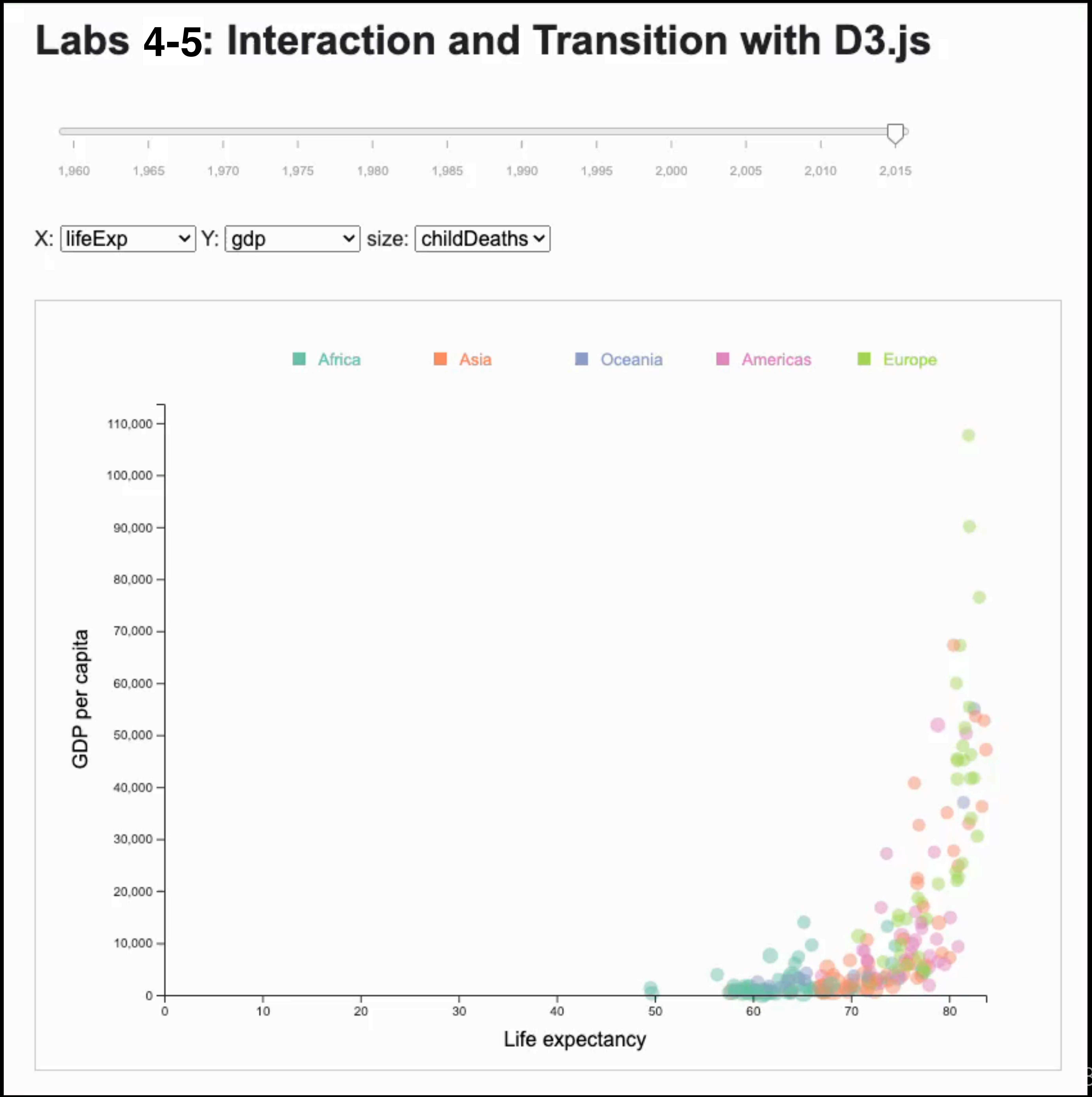

abstract figurative mechanistic

# Animation/Narrative

The introduction of time gives it a narrative structure

Carefully think about what you want to present along the timeline

# Wed Lab


Labs 4-5: Interaction and Transition with D3.js
1,960
1,965
1,970
1,975
1,980
1,985
1,990
1,995
2,000
2,005
2,010
2,015
X: lifeExp
Y: gdp
size: childDeaths
Africa
Asia
Oceania
Americas
Europe
110,000
100,000
90,000
80,000
70,000
60,000
50,000
40,000
30,000
20,000
10,000
0
GDP per capita
0
10
20
30
40
50
60
70
80
Life expectancy